JOURNAL of Marketing

STAMP: A Standardized Protocol and Web Application for LLM-based Text Classification

Journal:	*Journal of Marketing*
Manuscript ID	Draft
Manuscript Type:	New Submission
Research Topics:	Marketing Strategy, Marketing Analytics
Methods:	Archival analysis, Machine Learning/AI, Natural Language Processing, Text Analysis, Unstructured data analysis

SCHOLARONE™
Manuscripts

STAMP: A Standardized Protocol and Web Application for LLM-based Text Classification

Abstract

The authors develop and validate a systematic methodology for identifying and classifying marketing constructs in text corpora using large language models (LLMs). The Structured Taxonomy AI Measurement Protocol (STAMP) is a standardized, theory-grounded approach that features a multi-layer prompt architecture, a dual-LLM coding process, and a human adjudication process. The method's efficacy is demonstrated across three studies that utilize distinct text-based datasets spanning diverse marketing contexts, including brand communications, product reviews, and sales tactics. The STAMP approach achieves human-level performance (inter-rater reliability: Krippendorff's alpha > .67; and classification accuracy: F1 scores > .80). Compared to extant automated text classification methods such as dictionaries and supervised learning, STAMP offers a transparent, replicable, and scalable solution for analyzing marketing-related content by enhancing classification efficiency and contextual adaptability while maintaining theoretical rigor. This work bridges a critical gap between marketing measurement theory and artificial intelligence capabilities, providing researchers and managers with a standardized framework for analyzing text. A publicly accessible web application that implements the STAMP process accompanies this article (https://tinyurl.com/STAMPTextClassification).

Keywords: large language models (LLMs), textual analysis, construct classification, measurement theory

Peer Review Version

Introduction

The contemporary landscape of marketing research is marked by a paradox of abundance. Scholars now have unprecedented access to unstructured textual data in social media posts, online reviews, and customer service transcripts, among many other sources. This proliferation of digital text presents both a major opportunity and a significant measurement validity challenge for marketing researchers and practitioners. Historically, scholars have relied on manual human coding (typically using structured codebooks, thematic analysis, or inductive qualitative coding) for textual construct classification. Trained coders can discern cultural nuance, irony, contextual cues, and implicit meaning, ensuring high construct validity. However, textual data is often voluminous, making manual human-coding costly, prone to error, and unscalable. Furthermore, it is generally polysemic and context-dependent, requiring domain intelligence to ensure accurate interpretation and reliable classification (Carlson and Burbano 2025). As Bermejo et al. (2025) observe, manual coding is slow and expensive, and coder consistency can degrade over time.

Researchers have sought to overcome the limitations of manual human coding through automation. Early dictionary-based approaches (e.g., LIWC) and later machine learning (ML) classifiers enabled enhanced analysis of large textual datasets. These methods delivered substantial scalability but also led to a loss of measurement validity. Dictionary methods rely on fixed lexicons and provide consistent (deterministic) coding; however, they lack the contextual nuance required to interpret polysemous or context-dependent language, often leading to misclassification (Humphreys and Wang 2018). Supervised machine learning models (Naive Bayes, SVMs, neural nets) can capture more complex patterns with high predictive accuracy, but are generally considered “black boxes” that lack transparency, offer limited interpretability, require expensive labelled datasets, and may rely on spurious correlations unrelated to the

Peer Review Version

underlying construct. For example, a sentiment classifier may learn that certain punctuation patterns signal negativity, but without capturing the sentiment expressed. Post-hoc explainers like SHAP can identify which words influenced a model's decision, but they cannot answer why. (E.g., they might flag the word "great" as important while missing the sarcastic tone that flips its meaning). In adopting these automated approaches, researchers traded qualitative meaning for quantitative prediction, accepting a degradation in theoretical rigor in exchange for larger sample sizes. The result has been a persistent tension between measurement validity (i.e., depth, accuracy, and theoretical meaningfulness of measures) and scalability (i.e., the volume of data that can be analyzed).

Recently, large language models (LLMs) have emerged as a powerful alternative due to their ease of use, extensive general domain knowledge, chain-of-thought reasoning capabilities, and zero-shot and few-shot learning that reduces the need for voluminous labeled datasets (Schulhoff et al. 2025; Blanchard et al. 2025; Villarroel Ordenes et al. 2025). These massive generative transformers like GPT-4, Anthropic Claude, and Google's Gemini possess a latent expertise from training on internet-text which enables them to perform zero-shot classification with consistency and stamina far beyond any human coder. Recent evidence indicates that LLMs are shown to approach human expert-level coding quality while retaining the scalability of automated methods (Bermejo et al. 2025; Gilardi et al. 2023). These developments suggest that LLM-based coding may offer a viable and scalable alternative to large human coding teams. We present a detailed comparison of LLMs with extant text classification methods, such as dictionary and supervised ML, in Web Appendix A.

However, the integration of LLMs into scientific measurement introduces a new methodological challenge. Early applications of LLMs for text classification have found LLMs

Peer Review Version

to be highly sensitive to prompt phrasing, often suffering from a critical lack of theoretical grounding and methodological standardization (Ziems et al. 2024, Garvey and Blanchard 2025, Oh and Demberg 2025). This problem manifests in multiple forms: (a) zero-shot prompting which relies on the LLM's latent and often opaque definitions of constructs, (b) ad hoc prompt engineering, in which researchers iteratively adjust prompting techniques, wording, formatting, or instructions to obtain desirable results (similar to p-hacking), and (c) variations across LLM models or prompt contexts (Carlson and Burbano 2025; Blanchard et al. 2025). This leads to issues, such as replicability, where results are dependent on a specific LLM model or prompt phrasing (Abdurahman et al. 2024); demand effects, where iterative prompting can steer a LLM model toward a desired but unsubstantiated conclusion; and questionable measurement validity (Arora et al. 2024; Goli and Singh 2024; Lin 2027). For example, simple prompts like "Is this text adaptive selling?" can elicit judgments that reflect the model's internal representations rather than the researcher's conceptual definition, and minor linguistic variations can produce materially different outputs. Mu et al. (2024) demonstrate that small changes in wording, punctuation, or ordering of options shift accuracy by 10% on classification tasks. This fragility poses a threat to construct validity because the measurement may depend on subtle, non-theorized prompt features, undermining transparency and replicability.

Without a structured prompting framework, LLM-based text analysis may default to ad-hoc "prompt hacking," instead of rigorous and transparent measurement. We argue that progress in text-based LLM construct classification relies on the need for a standardized methodology that ensures adherence to established principles of measurement theory, transparency, and scalability, while achieving human expert level performance. To address the limitations associated with the use of LLMs for text analysis, we present herein a structured, theory-grounded LLM-based

Peer Review Version

coding process, referred to as the Structured Taxonomy AI Measurement Protocol (STAMP). STAMP integrates principles from psychometric measurement theory in marketing, (e.g., Churchill 1979; MacKenzie, Podsakoff and Podsakoff 2011) with recent insights from AI research, including natural language inference and ambiguity detection.

STAMP's design is built on three core components. First, it treats the LLM prompt as a structured measurement instrument rather than an ad-hoc query. In practice, the prompt serves as a digital codebook that explicitly embeds the researcher's theoretical constructs, inclusion/exclusion criteria, and exemplar texts. The prompt is formatted with a mix of structured markup (e.g., JSON-like or XML tags) and natural language instructions to clearly separate each section's content from its purpose. This structured format ensures that the model processes the construct definitions and coding rules as intended, grounding the prompt in construct validity principles akin to a traditional codebook (Churchill 1979; MacKenzie et al. 2011).

Second, STAMP employs a dual-LLM coding approach as a methodological analog to having multiple human coders for reliability. Two LLMs from different model families (e.g., Google Gemini, OpenAI GPT, and Anthropic Claude) independently code each text in parallel, providing redundant judgments on the presence of the construct. This approach enables an inter-model agreement check, comparable to inter-coder reliability in content analysis (Bojić et al. 2025; Krippendorff 2018). By leveraging heterogeneous LLMs, STAMP reduces idiosyncratic biases of any single model and produces auditable reliability metrics (e.g., Krippendorff's α) for the LLM-generated labels.

Third, STAMP treats inter-model disagreement not as a nuisance to be ignored or averaged out, but as a valuable diagnostic signal that guides refinement of the coding scheme. Rather than discarding cases where the two LLMs disagree, those instances are flagged

Peer Review Version

for human review and theoretical reflection. An auxiliary "judge" LLM is used to aggregate and summarize the points of disagreement across cases (akin to topic-modeling the contentious instances) to help the researcher identify why the models diverged. Such LLM-on-LLM disagreement often indicates borderline content or ambiguities in construct interpretation. By examining these cases, a human expert can adjudicate the correct classification and, importantly, update or refine the construct definitions and prompt instructions to resolve ambiguities going forward. This feedback loop utilizes disagreement as a prompt for clarifying the measurement instrument, ensuring that difficult edge cases inform stronger theory alignment (Bojić et al. 2025; Jayaweera and Dorr 2025). By formalizing prompt design and leveraging dual-LLM consensus, with human adjudication for AI disagreements, STAMP combines the interpretive rigor of human coding with the scalability of AI systems. In essence, STAMP synthesizes best practices from classical content analysis (e.g., clear codebooks, multiple coders, reliability checks) with the capabilities of modern LLMs, thereby embedding construct validity and reliability into the text classification process.

We validate STAMP across three diverse marketing contexts. In Study 1, we utilize employee-generated social media content (Instagram posts) in a luxury apparel brand environment characterized by nuanced language that requires specialized contextual knowledge. We code two theoretically grounded constructs: a "brand–product" dimension that captures references to classic, timeless luxury style, and a "brand–service" dimension that captures personalized service or product customization. In Study 2, we analyze Amazon.com product reviews to measure "customer recommendations". Consumer-generated product reviews are characterized by simpler and easier-to-understand language. Finally, in Study 3, we use a dyadic salesperson-customer interaction dataset to identify the use of "value confirmation" as a sales

Peer Review Version

tactic. Because dyadic interactions involve interdependent turns, accurate classification requires attention to evolving context, speaker intent, and conversational dependencies rather than isolated utterances. Together, these settings allow us to assess STAMP's performance across diverse linguistic and conceptual contexts.

Across the three varying marketing contexts, our study findings demonstrate that STAMP delivers reliability and validity levels that meet or exceed human benchmarks. STAMP achieves high inter-rater reliability (Krippendorff's alpha (α) > .67), satisfies construct validity criteria (significant Steiger's Z-test following multi-trait multi-method (MTMM)), and exhibits strong classification performance (F1 score > .80). To ensure the accessibility of STAMP for marketing scholars and industry analysts alike, we develop and provide a publicly accessible web application implementing the full STAMP workflow without requiring any installation or coding (https://tinyurl.com/STAMPTextClassification; see Web Appendix B for a user guide). This application helps translate STAMP's methodological innovations into a practical tool, lowering technical barriers for researchers and practitioners.

This research's contributions are thus threefold: First, we formalize an LLM-based text measurement protocol (STAMP) that embeds classical principles of construct development into a structured, multi-layer, codebook-style prompt architecture. By replacing ad-hoc prompting with a theory-aligned measurement instrument and incorporating a dual-LLM coding and human-adjudication process, STAMP reduces demand effects, leverages model heterogeneity, and produces auditable reliability metrics and high-quality labels with reduced effort and resources. Second, we demonstrate STAMP's generalizability across marketing contexts with markedly different linguistic and conceptual demands. Across three empirical settings, STAMP achieves human-level reliability and validity while operating at a fraction (only 1-2%) of the cost of

Peer Review Version

conventional human coding. Third, we offer STAMP as a standardized and accessible tool for text classification. The accompanying public web application enables scholars and practitioners to implement STAMP without technical setup, thereby enhancing transparency, replicability, and scalability. Collectively, these contributions integrate measurement theory with advances in LLM modeling, providing a rigorous, scalable, and transparent alternative to traditional methods such as dictionary, ML, and even a single LLM classifier used for text classification.

In the following sections, we situate the STAMP within the broader landscape of text classification approaches and outline the components of the structured protocol. We provide a step-by-step guide to the accompanying no-code, open-source application, which enables researchers to implement STAMP, including the dual LLM workflow and disagreement protocol, with no technical setup required. Finally, we evaluate STAMP's performance across three empirical studies. Lastly, we discuss the theoretical and practical implications of adopting a structured, multi-LLM measurement framework for text classification.

Literature Review: From Human Coding to Algorithm-Assisted Coding

To understand the different components of STAMP, we first position it within the evolution of textual classification methods. Successive approaches (i.e., manual human coding, dictionary-based, ML classifiers, and now LLMs) have each addressed and improved upon the limitations of their predecessors while also introducing new challenges. This progression highlights a persistent trade-off between measurement validity and scalability, a tension that STAMP is designed to mitigate.

Manual Human Coding as the Benchmark for Construct Validity

Peer Review Version

For much of the twentieth century, content analysis in marketing and the social sciences relied on manual human coding, in which trained coders iteratively developed and applied structured coding schemes (Berger et al. 2020; Krippendorff 2018). In this approach, validity is established through a systematic process of coder training, application, disagreement resolution, and refinement (Corbin and Strauss 1990). Researchers begin with a preliminary definition for the construct of interest (e.g., customer satisfaction or brand authenticity) and inclusion-exclusion criteria. Multiple independent coders then apply the definition and inclusion-exclusion criteria to a sample of text data. Disagreements among coders are not treated as errors but as diagnostic signals of ambiguity in the data or underspecified guidelines (Aroyo and Welty 2015; Jayaweera and Dorr 2025). Through recursive cycles of coding, discussion, and revision, the codebook becomes increasingly precise, resulting in a shared understanding of the construct and reliable evaluation criteria. This iterative, consensus-building process underpins the high construct validity of manual human coding.

Manual human coding, however, faces well-documented limitations. It is time-consuming, expensive, and difficult to scale. Even a single coder's judgments may vary with time and fatigue. Further, achieving acceptable inter-coder reliability often requires repeated training sessions. Recent studies confirm these weaknesses. For example, Bermejo et al. (2025) characterize manual expert coding as slow and expensive, and Bojić et al. (2025) document that the reliability of human annotators can be unstable over time as their attention and criteria shift. Consequently, while human coding offers strong validity, its lack of scalability constrains researchers to relatively small samples, raising concerns about reproducibility in large-scale textual datasets.

Dictionary Methods and the Pursuit of Scalability

Peer Review Version

The first major effort to scale textual classification emerged with dictionary-based methods. Tools such as LIWC (Linguistic Inquiry and Word Count) (Tausczik and Pennebaker 2010) and the Evaluative Lexicon (Rocklage et al. 2018) employ predefined word lists associated with psychological or semantic categories. By simply counting the occurrences of these words and their severity at the same time, researchers can automatically categorize text with perfect repeatability. These methods are easy to use, scalable (quickly processing millions of documents), require minimal human intervention beyond dictionary design, and are transparent (coding decision is deterministic and traceable to specific words).

However, the simplicity that affords ease of use and scalability also imposes important limitations. Dictionary-based methods treat language as literal and static, rendering them insensitive to context, pragmatics, polysemy, and linguistic ambiguity (Berger et al. 2020; Chakraborty et al. 2022; Humphreys and Wang 2018). A word such as "long" for example, may indicate duration in one context (e.g., a long meeting) and desire in another (e.g., I long for you), yet dictionary-based methods cannot distinguish these meanings. In short, dictionary methods prioritize reliability (each input yields the same output every time) at the expense of validity (capturing the true meaning). Despite their limitations, dictionaries remain widely used (even in leading marketing journals) because they offer a practical and scalable solution that overcomes the challenges of manual human coding. This underscores the need for scalable tools, even if it means accepting a blunt proxy for the construct of interest and further highlights the need for scalable methods that accurately capture the contextual nuance essential for valid measurement.

Machine Learning (ML) Classifiers and the Challenges of Interpretability

The next major development in text classification came from machine learning approaches, including supervised algorithms such as Naïve Bayes and support vector machines,

Peer Review Version

as well as later neural network models, including convolutional architectures and fine-tuned BERT. These methods improve on dictionary methods by learning patterns from labeled examples. Given a sufficiently large, labeled dataset (i.e., texts hand-coded by humans), an ML classifier can generalize and apply labels to new data. When compared to stand-alone dictionaries, ML classifiers achieve higher accuracy and can handle subtle features such as word combinations and syntax.

Despite the advances, ML classifiers introduce important limitations. Many operate as "black boxes" (Lundberg and Lee 2017), making it difficult to determine whether their predictions truly capture the intended construct or are influenced by correlations unrelated to it. For instance, a model might label a product review as negative, not because it has learned and interpreted the sentiment itself, but rather by spurious associations such as punctuations and emoji usage. Explainable AI tools can highlight influential features such as words, n-grams, etc., but knowing which words influenced the model is not equivalent to understanding the contextual or semantic reasoning behind it. Moreover, ML classifiers need large, task-specific labeled datasets, making them costly to develop and difficult to adapt to new constructs or domains. In short, although ML classifiers offer improvements in predictive performance and flexibility over dictionary methods, they do not fully address the construct validity problem. Further, they also introduce issues of interpretability and agility. As noted in prior work, ML-based text classification prioritizes out-of-sample accuracy as the primary evaluation criterion, with less attention to whether the model's definitions align with theory (i.e., construct validity) (Berger et al. 2020; Humphreys and Wang 2018). Consequently, the gap between measurement and theory has widened in the age of ML. Researchers have gained the ability to analyze much larger

Peer Review Version

corpora, but often at the risk of relying on models that may be empirically accurate yet theoretically misaligned.

In summary, the evolution of text-classification methods reveals a persistent set of trade-offs. Human coding provides depth and strong construct validity but lacks scalability. Dictionaries provide speed and consistency but can misinterpret linguistic context. ML methods provide better accuracy and adaptability yet operate as a black box and can diverge from theoretical definitions. Notably, none of the methods simultaneously delivers high validity, high scalability, and high transparency. A review of prior studies (see Web Appendix C) also highlights the fact that studies using text-based constructs often fail to report multiple validity checks, an issue frequently overlooked in prior research.

Using a Single LLM as a Text Classifier.

Recent research suggests that, with proper guidance, LLMs may help reconcile the tensions of previous methods (Arora et al. 2024; Bermejo et al. 2025; Blanchard et al. 2025). Thanks to vast language pre-training, LLMs combine the contextual sensitivity of humans with the consistency and efficiency of machines. Existing research offers insights into LLM-based text analysis, but does not provide a comprehensive measurement protocol. Bermejo et al. (2025) demonstrate that GPT-3.5/4 used in zero-shot mode to code news articles can outperform student coders and deliver substantial efficiency gains. However, their focus was on accuracy against human labels; they did not incorporate a construct validity framework and essentially treated labels as given truths or theory-driven prompting. Chakraborty et al. (2025) showed the value of hybrid AI-human approaches in a predictive context (sales hiring), but their approach did not formalize a general coding protocol for constructs. Gilardi et al. (2023) document that LLMs can be inexpensive and reliable annotators for tasks such as content moderation, yet their evaluation

Peer Review Version

centers on performance and cost rather than on ensuring alignment between model outputs and theoretical definitions. Carlson and Burbano (2025) emphasize the importance of prompt design and robustness checks, illustrating that variations in prompts or models can undermine substantive research conclusions.

STAMP addresses the known challenges of using a single LLM for text classification by providing a standardized prompt and an iterative testing mechanism based on model disagreement. Rather than averaging across large sets of prompts, STAMP resolves prompt sensitivity through theoretically grounded construct boundaries. Bojić et al. (2025) highlighted the importance of inter-rater reliability in LLM outputs; however, their work does not incorporate reliability diagnostics within the validation loop. STAMP extends this insight by using inter-model disagreement as a signal of ambiguity that guides refinement. Finally, Goli and Singh (2024) demonstrate that LLMs, when used as simulated respondents, exhibit systematic biases such as impatience in the model's behavior, underscoring the need for rigorous validity checks. STAMP addresses this through a multi-model agreement and structured prompting. In sum, prior studies provide encouraging evidence that LLMs can be powerful tools in text analysis, but they also reveal important vulnerabilities. Outcomes depend heavily on the prompt's fidelity, and LLMs' judgments can deviate from theoretical constructs. STAMP synthesizes these insights into a unified, theory-driven protocol that integrates structured prompts, disagreement diagnostics, and human oversight. By doing so, it ensures that construct validity is embedded in the coding process. We provide a detailed comparison of single LLM-based Text Classification and STAMP in Web Appendix D.

STAMP Framework for Valid and Scalable Text Classification

Peer Review Version

The STAMP framework comprises three components. First, it treats the LLM prompt as a measurement instrument rather than an ad hoc query, embedding construct definitions, inclusion–exclusion criteria, and illustrative examples in a structured, codebook-like format. This design draws on fundamental principles of measurement theory, as the prompt is developed and validated, much like a multi-item scale, in order to ensure construct validity. Second, it employs multiple LLMs from distinct model families as independent coders, enabling cross-model validation of classifications. By having two heterogeneous models work in parallel, STAMP mirrors the practice of using two independent human coders to establish reliability (Krippendorff 2018). Third, STAMP interprets inter-model divergence and accompanying reasoning traces as diagnostic signals, rather than instances of disagreement, to guide targeted human review.

Prompt as a Structured Measurement Instrument

STAMP addresses the prompt-sensitivity problem by reconceptualizing the prompt as a multi-layered LLM measurement workflow rather than a one-off query. It draws on Churchill's (1979) paradigm for construct measurement, which emphasizes careful specification of the construct domain, generation of measurement items, and iterative purification to embed theory into prompt design. STAMP maps these steps onto a three-layer prompt that guides the LLM's behavior. Specifically, the prompt consists of three essential layers—definition, inclusion and exclusion criteria, and guided examples. In the STAMP web application, these layers are delineated using a markup-style syntax (inspired by XML tags) to clearly separate each component of the digital codebook prompt (Anthropic, 2025; Google Cloud, 2025). This structured, machine-readable format reduces the model's misinterpretation of the guidelines and supports reproducibility by enabling the LLM to output structured JSON results that mirror the prompt tags, thereby facilitating automated auditing of the coding process. Essentially, the

Peer Review Version

STAMP prompt serves as a digital measurement instrument that can be refined, standardized, and reused across studies to enhance replicability. This shared understanding between humans and LLMs facilitates scalability and verification. Next, we discuss the three essential prompt layers. We use the construct of "personalized service" (from Study 1) as an illustrative example as we discuss the STAMP prompt.

Layer 1: Construct Definition. The first layer of the prompt is similar to a Zero-shot classification, wherein we provide a formal definition of the focal construct that is grounded in relevant literature. Zero-shot classification is the canonical starting point for evaluating an LLM's latent understanding (Schulhoff et al. 2025) and is therefore the baseline in STAMP's workflow as well. However, prior work indicates that zero-shot classification performance is often construct-dependent. In other words, LLMs can match or exceed humans in classifying concrete or well-known constructs (e.g., sentiment), but often diverge across models and underperform in more abstract or ambiguous categories (Ziems et al. 2024; Bojić et al. 2025). Further, simply pasting a human codebook with all relevant details into a zero-shot prompt is found to be insufficient, as classification performance improves when tasks are decomposed and the context window is more tightly defined (Hohenwalde et al. 2025). So rather than simply naming a category, the prompt explicitly defines the construct in clear, accessible language to ensure a shared understanding. For example, a prompt might define "personalized service" as "content showing that a brand customizes products or services to an individual client's needs, desire, lifestyle, or preferences.." As Churchill (1979) notes, specifying the domain of the construct is a critical first step in valid measurement. STAMP operationalizes this by anchoring the LLMs to the researcher's exact construct definition, ensuring that the LLM begins reasoning from an expert-like foundation and internalizes a definition before measuring, rather than

Peer Review Version

allowing the model to rely on its own latent understanding. This focuses the model's attention on the theoretically relevant features of the text and reduces the influence of irrelevant associations learned during pretraining. A precise initial definition helps ensure that subsequent LLM judgments are about the intended construct and not drifting toward related but distinct ideas. At the same time, a definition-only zero-shot prompt preserves space for the LLMs to express their initial understanding or assumptions without overly constraining them to the researcher's perspective, which can be informative during the exploratory phases of construct development (MacKenzie et al. 2011).

Layer 2: Inclusion/Exclusion Criteria. The second layer enumerates explicit decision rules for what should be included as the construct (inclusion criteria) versus what should be excluded (exclusion criteria), even if related. In traditional scale development, researchers generate candidate items that capture the construct and articulate what the construct is not, thereby ensuring discriminant validity. STAMP mirrors this logic by providing the LLM with concrete positive and negative coding rules. The inclusion/exclusion criteria are derived from literature and expert judgment, serving as a codebook with decision rules that mirror human-coder training. For instance, in actual salesperson social media posting practices, what counts as "personalized service"? Does a call-to-action (CTA) to set up a sales consultation count, or is it simply a generic CTA or standard sales pitch? Does stating a product feature that could be adapted to an individual count, or must the personalization be explicitly stated? These decisions require understanding the data context and operationalizing it consistently with the theory being tested. Therefore, an inclusion criterion might be "Explicitly personalized experience—describes a one-to-one, private, or individualized shopping/delivery interaction (e.g., 'private Zoom fitting,' 'one-on-one styling session,' 'house call tailoring,' 'personal wardrobe consultation')." An

Peer Review Version

exclusion criterion might be "Standard retail variety—ONLY discussion of unique fabrics, wide product ranges, or other features that do not involve tailoring to the individual." These criteria clarify the scope of the construct and prevent concept drift during classification. By operationalizing aspects of discriminant validity in the prompt, Layer 2 creates guardrails that keep the model from conflating the target construct with adjacent concepts. For example, a naive model might confuse service quality with product quality when assessing "satisfaction" in a review; an exclusion criterion can explicitly instruct the model to ignore statements about shipping or customer service if the construct of interest is product satisfaction. This structured guidance parallels the detailed instructions provided to human coders and promotes consistency in how the construct is applied across texts.

Layer 3: Examples and Chain-of-Thought Rationales. The third layer provides prototypical examples of text for each category, accompanied by the correct label and a rationale explaining the coding decision. The examples consist of typical cases and edge cases serving as a form of few-shot training within the prompt. The inclusion of rationales ("explain why this text is or isn't an example of the construct") further enhances learning and forces the LLM to connect the definition and criteria to real text instances, thus improving the LLM's reasoning transparency (i.e., the model is taught not just what the answer is, but also why that answer is correct). This step parallels the "purification" stage in Churchill's (1979) framework, in which initial measurement instruments are refined and validated through applied examples. By presenting prototypical cases along with the reasoning behind their classification, STAMP guides the LLM in how to apply the measurement logic rather than merely assigning labels. Providing examples with explanations is known to boost model performance on classification tasks by giving concrete reference points. For instance, to implement the inclusion and exclusion rules for

Peer Review Version

concession, a structured example with embedded chain-of-thought reasoning based on the inclusion and exclusion criteria can be:

Statement to detect: "Browse unique-one-of-a-kind fabrics along with the last pieces of our best-selling fabrics to complement your wardrobe."

{

"reasoning": "inclusion_criteria [] and exclusion_criteria [1]; explanation: The statement only discusses unique fabrics and product variety without mentioning any personalization, customization, or individual tailoring aspects.",

"tactic_prediction": 0

}”

The reasoning component is central, as it communicates the desired interpretive process and helps align the model’s internal reasoning with the researcher’s theoretical specification. This “glass box” approach, where the AI’s reasoning is made explicit, aligns with evidence that transparency in algorithmic reasoning improves human–AI collaboration and oversight (Balakrishnan et al. 2025). Prior work also shows that providing examples can stabilize LLM outputs, although performance remains sensitive to the content and wording of the prompt (Carlson and Burbano 2025). Well-crafted exemplars and reasoned justifications help mitigate this sensitivity by providing a template for how the model should interpret and evaluate new texts. As the LLM’s decision process is shaped by implicit and explicit examples and reasoning, its eventual outputs are easily traced back to these known reference cases. In essence, Layer 3 enhances transparency and reliability: the model is expected to produce similar step-by-step rationales when coding unseen data, which allows human adjudicators to trace and audit the

Peer Review Version

model's decision-making.

Thus, the STAMP workflow begins with a definition-only prompt and subsequently adds inclusion/exclusion criteria and examples, while also capturing chain-of-thought rationales for internal auditability (Schulhoff et al. 2025; Balakrishnan et al. 2025). Together, the three layers (i.e., definition, inclusion/exclusion criteria, and guided examples) form a comprehensive and structured prompt that acts like an expert, enhancing the content validity of the underlying construct. This transformation converts an otherwise vague zero-shot prompt into a structured, codebook-based measurement instrument. By specifying clear construct definitions, boundaries, and exemplars, STAMP directly mitigates the prompt-sensitivity problem, reduces reliance on the model's idiosyncratic interpretations, and mirrors established best practices in measurement development (Churchill 1979; MacKenzie et al. 2011). In this sense, the STAMP prompt substantially enhances the consistency and human alignment of LLM outputs, and thus can be used as a reusable instrument or protocol for data annotation, one that can be refined over time and applied across studies to achieve reproducible results.

Multiple LLMs as Independent Coders

A crucial design choice in STAMP is the use of heterogeneous LLMs for the dual-coding process. Rather than relying on two instances of the same model, STAMP identifies and pairs two LLMs that have similar capabilities[1] but belong to diverse development families (e.g., Google Gemini, OpenAI GPT, and Anthropic Claude. The classification process is executed in parallel across the two distinct LLMs. For each text unit in the dataset, the same prompt is sent to both models (e.g., OpenAI's GPT-4.1 and Google's Gemini Flash 2). The protocol integrates

[1] We use Massive Multitask Language Understanding (MMLU) to assess model capabilities, as it is a commonly accepted benchmark for assessing LLMs' knowledge and reasoning (Hendrycks et al. 2021).

Peer Review Version

chain-of-thought reasoning by instructing each model to first "think step-by-step" and explain its rationale before providing a final classification. This practice enhances transparency, allowing researchers to audit the model's decision-making process. Because these systems differ in architecture, training corpora, and inductive biases, they are unlikely to share the same idiosyncratic errors or blind spots. Convergence across heterogeneous models, therefore, provides stronger evidence that a given classification reflects the underlying construct and the clarity of the prompt, rather than any model-specific quirk (i.e., cross-model validation of the coding). Using models from competing developers enhances coverage (i.e., utilizes a broader range of training data to train the models), reduces skew, enhances fairness, and improves generalization (akin to human experts with diverse perspectives). By requiring agreement between two architecturally distinct LLMs, STAMP ensures that classifications are robust and not an artifact of one model's specific training or failure modes. Empirical findings support this approach: advanced LLMs can serve as reliable coders in content analysis, with GPT-4.1 even matching or exceeding human inter-coder reliability on certain tasks (Bojić et al. 2025). Each LLM acts as an independent coder with its own perspective; therefore, when they independently produce the same label and similar rationale, this agreement leads to stronger confidence that the classification is theoretically aligned and robust. We present a comparison of LLM models from diverse families across ability and cost in Web Appendix E, Table W1 and W2.

By using two models in parallel, STAMP harnesses this reliability and simultaneously introduces a new safeguard, as disagreement between the models becomes a signal to investigate. This design closely parallels the established practice of having two independent human raters for qualitative coding to ensure consistency (Krippendorff 2018). Prior work shows that when annotators (or models) disagree, it is often a diagnostic signal of ambiguity or uncertainty in the

Peer Review Version

data or definitions, not merely random noise. For example, Turpin et al. (2023) found that some LLMs exhibit sycophantic explanations, generating plausible rationales for almost any answer in order to appease the user, even if the answer is essentially a guess or error. A single LLM might thus rationalize its own misclassification, making it appear confident. By contrast, it's unlikely for two different LLMs to converge on the same wrong answer with the same justification. Disagreement across models, therefore, is highly informative. In practice, we observe that model disagreement frequently arises in cases where one model is confidently in one perspective and the other model flags the alternative view. This creates a system of checks and balances between the AI coders, akin to having two distinct experts independently review each case. Only when distinct LLMs independently converge on the same label and provide consistent rationales do we have high evidence that the classification is valid. In all other cases, the inter-model divergence is treated as a cue for closer examination.

Disagreement as a Diagnostic Signal – An Iterative Reliability Protocol.

In qualitative content analysis, disagreements between human coders are used to reveal conceptual ambiguity and to refine the coding scheme. STAMP similarly leverages LLM disagreements as a diagnostic indicator of ambiguity, by interpreting disagreement between heterogeneous LLM coders as an epistemic signal that either the text permits multiple interpretations or the construct definition and criteria require clarification. This view is consistent with emerging work in annotation science, which suggests that disagreement frequently reflects genuine content ambiguity or underspecified guidelines rather than coder mistakes (Aroyo and Welty 2015; Jayaweera and Dorr 2025). The CrowdTruth framework, for example, argues that variability in labels can expose important conceptual uncertainties and should not be discarded as

Peer Review Version

noise. Therefore, rather than simply discarding conflicting LLM answers, STAMP operationalizes them to identify areas where the codebook might be improved.

To make disagreement actionable, STAMP introduces a summarizer model as a third component in the coding loop. This tertiary LLM examines the rationales produced by the two primary LLM coders and synthesizes a brief report on the reasons for their disagreement. The summarizer identifies recurring themes underlying the divergence (e.g., ambiguous phrasing in the text, overlapping boundaries between related constructs, or consistent misinterpretations of contextual cues). In effect, this step converts raw inter-LLM divergence into a structured diagnostic report, highlighting where refinement of construct definitions, inclusion–exclusion rules, or exemplars may be needed. Using the summarizer's report, a human researcher can then intervene to adjust the coding prompt. For example, a theme identified for the "personalized service" was Consultation/Appointment Offers vs. Actual Personalization. The models disagreed on whether offering consultations constitutes personalized service. For instance, AI1 correctly identified the statement as only providing contact information and not any explicit evidence of customization. However, AI2 answered incorrectly by assuming that offering consultation automatically equals personalized service. Herein, the word "consultation" alone does not inherently indicate customization. Such disagreements stemming from conceptual ambiguities can prompt updates to the construct definition or criteria (revisiting Layers 1 and 2 of the prompt), whereas disagreements attributable to systematic model-specific behavior might lead to swapping out one of the models or otherwise reweighing their votes. If a disagreement legitimately reveals multiple interpretations of the construct, the researcher might decide to broaden/redefine the construct or add a new category to the coding scheme. Thus, high-disagreement cases become opportunities to refine theoretical definitions rather than anomalies

Peer Review Version

to be discarded. After making adjustments, the refined prompt is re-run on the dataset (or on the flagged cases), and the cycle repeats. This loop continues iteratively until disagreements are minimized and the construct boundaries are well-stabilized. As Uma et al. (2022) note, distinguishing between noise and systematic sources of disagreement is crucial for developing reliable measures, and STAMP operationalizes this principle by identifying whether divergence arises from unclear coding instructions, model-specific features, or genuinely ambiguous items. As STAMP cycles through disagreement, diagnosis, and refinement, it mirrors the consensus-building process used in human coder training, however, operating at machine scale.

Reliability and The Path to Validity

Achieving high inter-coder reliability has long been a basic requirement in content analysis, as unstable coding cannot yield a valid measure. This connection between reliability and validity is well established. Gilardi et al. (2023) show that higher inter-coder agreement is associated with greater accuracy relative to expert-coded ground truth. STAMP's disagreement resolution protocol is designed to reach high reliability by iteratively clarifying construct definitions and resolving sources of ambiguity. Specifically, when inter-LLMs agreement is high (Krippendorff's $\alpha > .80$), it indicates that the construct has been operationalized with sufficient precision for independent coders. Furthermore, consensus across LLMs within STAMP serves as evidence that coding decisions are no longer driven by underspecified instructions or model-specific heuristics but by the construct's substantive definition.

The final step in the STAMP workflow formalizes this logic. After an initial coding pass, inter-LLM reliability metrics are automatically computed. When reliability falls below accepted thresholds (Krippendorff's $\alpha < .67$), a summarizer model analyzes the rationales of the

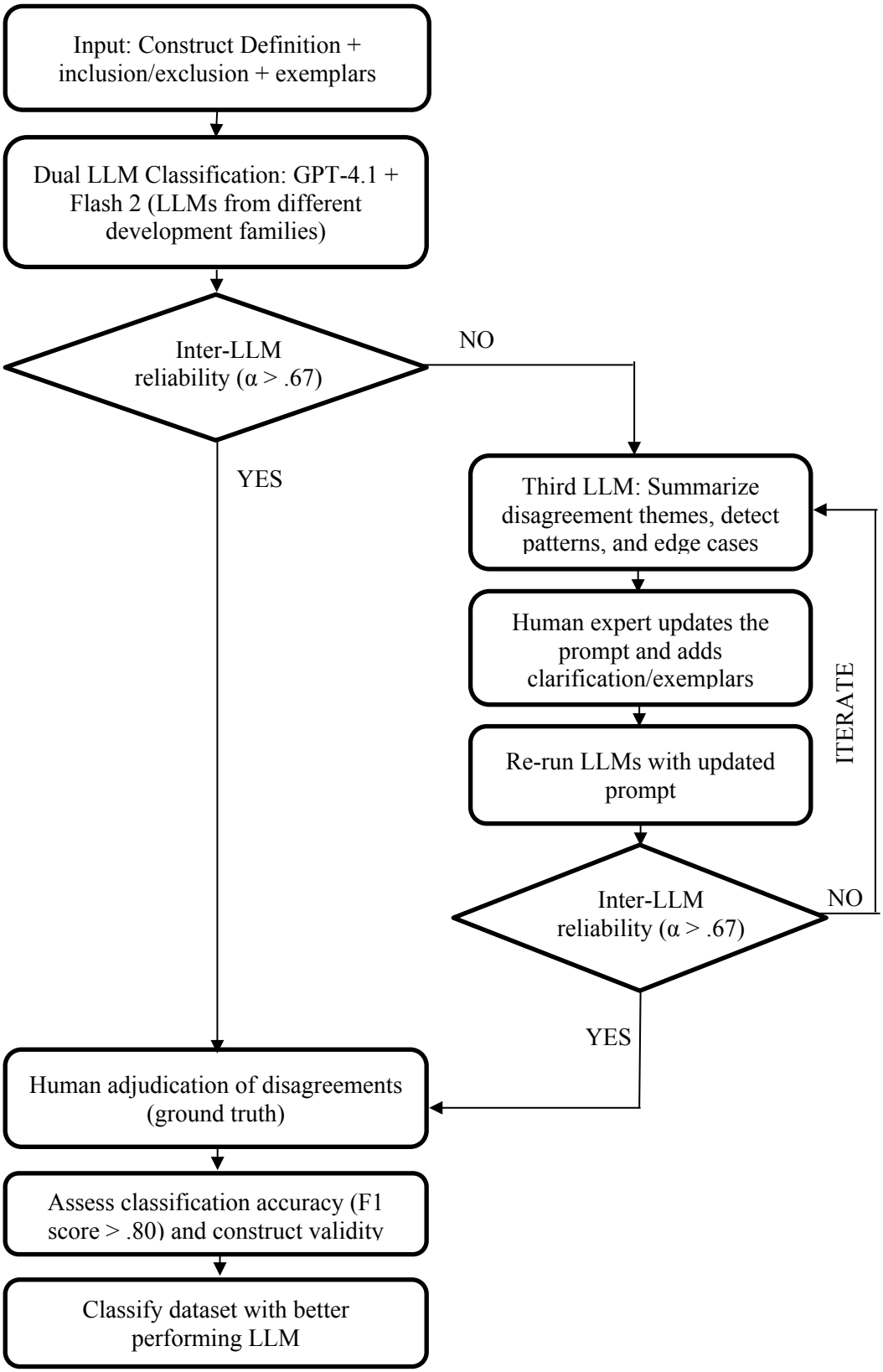
Peer Review Version

disagreeing LLMs and identifies themes underlying the divergence. The key themes are reviewed by an expert human researcher, who refines the prompt based on construct definition, inclusion and exclusion criteria, and/or exemplars before repeating the classification process. This iterative process continues until a satisfactory level of inter-LLM reliability is achieved (Krippendorff $\alpha > .67$). Remaining instances of disagreement are flagged for efficient human adjudication, thus using human expertise precisely where it is needed the most (i.e., for resolving edge cases that automated systems find challenging, and for establishing ground truth). This yields a dataset that combines the scale of automated classification with the conceptual rigor of expert judgement. By resolving disagreements through structured diagnostics and human oversight, STAMP transforms a sensitive prompt into a stable, theory-aligned instrument, ensuring that the final dataset is of gold-standard quality and anchored in ground truth, while leveraging AI for the majority of classifications. This approach achieves both rigor and scale.

STAMP also offers agility unmatched by fine-tuned ML classifiers. For ML classifiers any change in the construct definition or boundary condition requires new labels and model retraining. In contrast, STAMP refines the prompt, not the model. Updates to construct definitions or coding rules can be incorporated immediately, which enables researchers to iteratively adjust and test construct boundaries without the costs of retraining or recoding. Such flexibility is valuable for emerging constructs and dynamic domains such as social media, where conceptual refinement is ongoing. As a whole, STAMP's dual-LLM reliability checks, heterogeneous model design, and fast turnaround ensure both high reliability and validity. Figure 1 summarizes the full STAMP workflow. We next introduce our no-code, accessible web tool, which utilizes STAMP for text classification.

Peer Review Version

Figure 1. Graphical Representation of the STAMP Workflow



STAMP Web Application

To facilitate the broad accessibility of STAMP, we developed a publicly available web application that encapsulates the entire text classification workflow (https://tinyurl.com/STAMPTextClassification). This user-friendly tool will enable researchers and practitioners to apply the standardized LLM-based construct classification to textual datasets without requiring custom code. Below, we summarize the web application workflow and offer a step-by-step user guide in video format at https://figshare.com/s/7442f7dc932ae41f4126?file=59126732. For additional details, refer to Web Appendix B.

Setup: To use the web app, create an account with openrouter.ai, purchase credits, and enter the user API key into the STAMP interface to enable personal LLM access.

Data Upload: Prepare and upload the CSV data file.

STAMP Prompt Construction: Enter details for each layer of the STAMP structured prompt (i.e., construct definition, inclusion criteria, exclusion criteria, and examples). We format the prompt using a structured, markup-style syntax (inspired by XML) to clearly delineate each section. For example, the prompt might explicitly label sections with tags like <definition>...</definition>, <inclusion_criteria>...</inclusion_criteria>, <exclusion_criteria>...</exclusion_criteria>, and <examples>...</examples>. We suggest starting with a definition-only zero-shot prompt to explore the construct in the data context.

STAMP Classification: Run STAMP Classification by first selecting two LLMs (default: GPT-4.1 and Gemini Flash 2). Review inter-LLM agreement statistics (Krippendorff's α and percent agreement) and summary of disagreement cases. After achieving acceptable reliability, adjudicate remaining disagreement themes that have been flagged for human review.

Peer Review Version

Revised STAMP Prompt: If the initial inter-LLM agreement is below the standard threshold ($\alpha < .67$), use the app button to input the third-LLM summarized disagreement themes and re-run STAMP with the revised full prompt. Continue until agreement between the dual-LLM models meets or exceeds the standard threshold (Krippendorff's $\alpha > .67$).

Results and Export: Download the classified dataset and accompanying performance statistics. The LLM produces its answer in a JSON or similarly structured format that mirrors the structure of the prompt. The parser-friendly output ensures that the LLM's response (which might include its classification decision and reasoning) can be easily extracted without errors.

Figure 2 is a graphical representation of the STAMP app workflow, using "personalized service" as an illustrative example. We present the associated full-structured prompt and disagreement themes for "personalized service" in Web Appendix F.

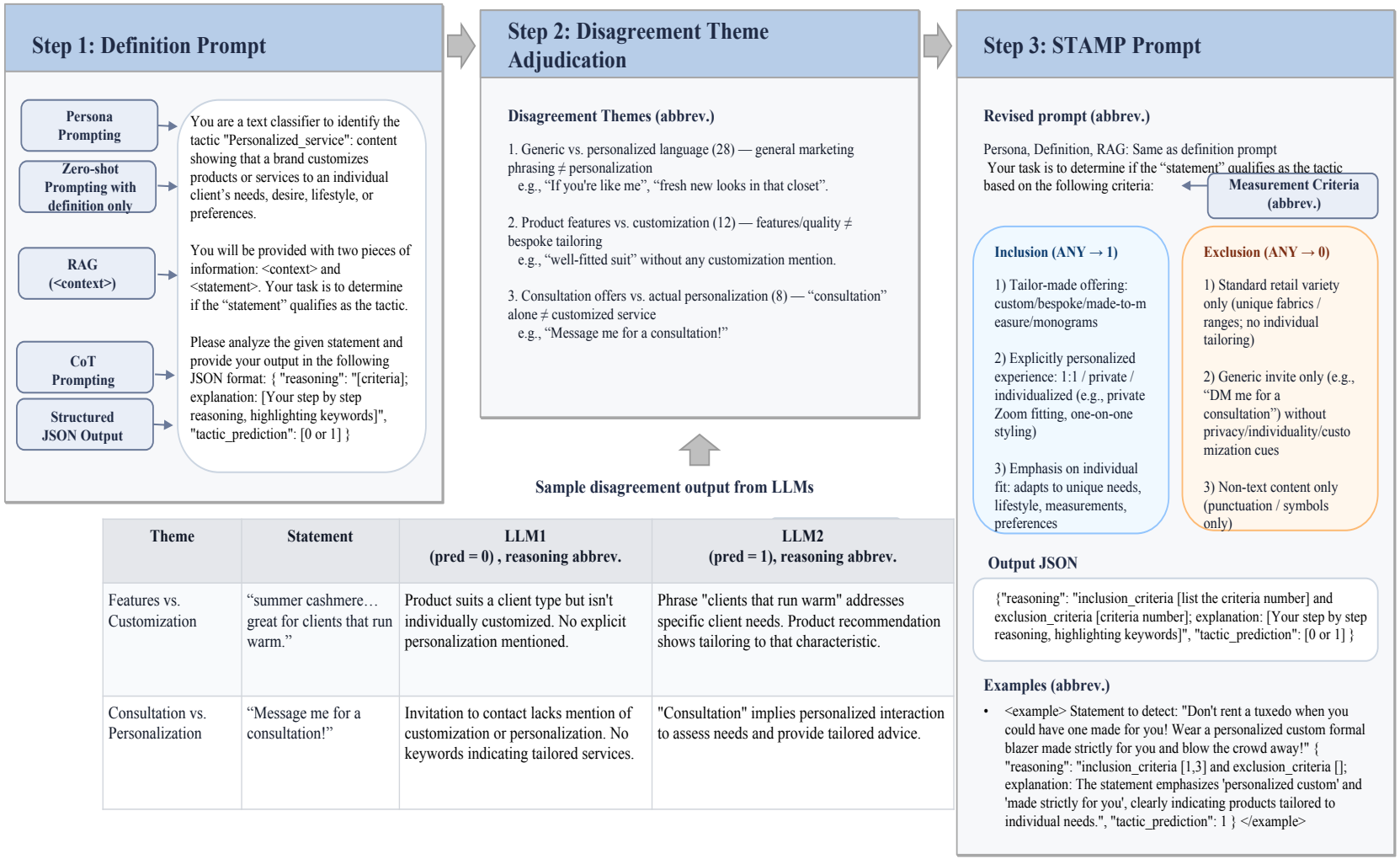
Empirical Validation of STAMP

For empirical validation, we evaluate STAMP's performance on three criteria: inter-LLM reliability (Krippendorff $\alpha > .67$), classification accuracy (F1 score > .80), and construct validity (convergent and discriminant) across three marketing contexts that differ in linguistic complexity and communicative intent.

In Study 1, we utilize a dataset of 1,000 social media posts randomly sampled from 22,583 Instagram posts created by sales professionals at a global luxury apparel brand. We examine the posts' text to assess whether salesperson-generated content that is aligned with specific brand image dimensions leads to better sales outcomes (Sirianni et al. 2013). We selected two distinct but related brand dimensions as targets for classification, based on written responses from both employees and customers describing what defined the focal brand. First,

Peer Review Version

Figure 2. **Graphical Representation of the STAMP App Workflow (Construct: Study 1 - Personalized service)**



Theme	Statement	LLM1 (pred = 0) , reasoning abbrev.	LLM2 (pred = 1), reasoning abbrev.
Features vs. Customization	"summer cashmere… great for clients that run warm."	Product suits a client type but isn't individually customized. No explicit personalization mentioned.	Phrase "clients that run warm" addresses specific client needs. Product recommendation shows tailoring to that characteristic.
Consultation vs. Personalization	"Message me for a consultation!"	Invitation to contact lacks mention of customization or personalization. No keywords indicating tailored services.	"Consultation" implies personalized interaction to assess needs and provide tailored advice.

Classic Luxury Style (i.e., content that emphasizes the brand's timeless, refined, and heritage-based product aesthetic), and second, Personalized Service (i.e., content that indicates customization of products or services to meet individual client needs. As this context is characterized by subtle aesthetic cues and sophisticated brand messaging, we treat it as a high domain-knowledge context, suitable for testing STAMP's ability to handle nuance and implicit meaning.

In Study 2, we use a dataset of approximately 600 sentences drawn from 100 recent Amazon reviews of car vacuum cleaners. Here, we classify the presence of Recommendations (i.e., explicit endorsements (warnings) that encourage (discourage) others from purchasing the product). Recommendations are theoretically and managerially important because they are associated with downstream marketing outcomes, such as helpfulness and product sales. This context is characterized by relatively direct and explicit consumer language; therefore, we consider it a low domain-knowledge context.

In Study 3, we extend STAMP's testing to a dyadic setting using a dataset of sales negotiations that happen over chat. The dataset consists of 50 sales negotiations and includes 1000 salesperson turns. Our focal construct is "value confirmation," a sales tactic in which sales agents restate product benefits to emphasize value and address price objections. Unlike the largely monologic texts in Studies 1 and 2, dyadic interactions are informal, fragmented, and interdependent, with tactics often spanning multiple communication turns. This setting poses a stringent test for STAMP, as the classifier must distinguish general product descriptions from strategically timed negotiation moves embedded in a back-and-forth exchange

Across all studies, we implement STAMP by specifying construct definitions, inclusion/exclusion criteria, and exemplars. We then iteratively apply two heterogeneous LLMs

Peer Review Version

(GPT-4 and Gemini Flash 2) until inter-LLM reliability reaches an acceptable level (Krippendorf alpha > .67). This approach is analogous to conventional human coder training in which coders classify, discuss disagreements, and revise guidelines under expert supervision. Importantly, STAMP leverages LLM-generated step-by-step reasoning to comprehend the reasoning behind labels, thereby enhancing transparency and supporting the refinement of the codebook (Wang et al. 2024; Schulhoff et al. 2025; Anthropic 2025; Google Cloud 2025). The resulting codebook[2] is then applied by both a trained research assistant (RA) and crowd workers, enabling assessment of cross-group reliability.[3]

For all three studies, ground-truth labels[4] were established through a rigorous two-stage process that combined expert judgment with a majority vote aggregated from crowd workers. First, multiple crowd workers labeled each item, and the majority vote was used to derive a consensus label from the crowd. Prior literature suggests that aggregating four to five non-expert judgments per item can approximate expert label quality (Snow et al. 2008). Second, a trained graduate student served as the coder and provided an independent set of labels, consistent with best practices in construct measurement and content analysis (Artstein and Poesio 2008; MacKenzie et al. 2011; Krippendorff 2018). These human-coded labels establish benchmarks for human-human reliability and form the basis for evaluating LLM-based approaches.

To assess predictive validity, a senior domain expert adjudicated all remaining disagreements among the LLMs (GPT-4.1 and Gemini Flash 2) and human coders (trained RA

[2] For study 3, we pre-registered our STAMP prompt and coding protocol for concession tactic provided to humans. See [https://aspredicted.org/fb3kq8.pdf]
[3] In the recommendation dataset, RA coding did not meet reliability thresholds; therefore, the senior researcher's coding was substituted for expert–expert reliability calculations. All results are reported in the "Results" section.
[4] We generate ground truth with the help of humans to assess STAMP's credibility and performance for a no-LLM counterfactual. However, this step is redundant if practitioners use STAMP for text classification.

Peer Review Version

and crowd majority vote). These expert adjudications served as the ground truth, addressing known concerns about LLM-LLM evaluation bias (Laurito et al. 2025; Zheng et al. 2023) and ensuring that final classification reflected contextual and theoretical expertise. Importantly, neither the LLMs nor the human coders had access to these adjudicated labels during the coding process. Collectively, this design allows us to benchmark STAMP against human coders, assess its reliability and accuracy, and evaluate its capacity to deliver valid construct measurements at scale. For full structured prompts and disagreement themes associated with all study constructs, see Web Appendix F.

Results

Across the three studies, our results demonstrate that STAMP meets reliability thresholds (Krippendorff's $\alpha > .67$), achieves accurate classification (F1 score > .80), and establishes construct validity (both convergent and discriminant).

Inter-LLM Reliability.

To assess inter-LLM reliability, we compare the performance of a definition-only prompt with full structured STAMP prompt for all three studies. To transition from a definition-only to a fully structured STAMP prompt, we utilize model disagreements that ensure the final STAMP prompt addresses real classification challenges rather than theoretical ones. The systematic analysis of disagreement patterns provides transparency into why certain classification decisions are challenging, allowing researchers to refine construct boundaries with empirical evidence rather than relying solely on intuition. This iterative process continues until inter-LLM reliability reaches acceptable levels (Krippendorff's $\alpha > .67$), ensuring the final prompt produces consistent and valid classifications across different model architectures. See Web Appendix G (definition-

Peer Review Version

only) and Web Appendix F (full structured STAMP) for the number and type of disagreement themes associated with each prompt for "personalized service."

In Study 1, as expected, the definition-only prompt yielded inter-LLM agreement of .37 (Classic Luxury Style) and .55 (Personalized Service). However, with a full-structured STAMP prompt, which additionally includes inclusion/exclusion criteria and exemplars, the inter-LLM reliability reached acceptable levels (.77 for Classic Luxury Style and .87 for Personalized Service). A similar pattern repeated for studies 2 and 3. In Study 2, the inter-LLM agreement for recommendations increased from .55 (for definition-only) to .69 (for the STAMP prompt). In Study 3, the inter-LLM agreement for Value Confirmation was .67 (for the definition-only prompt), which increased to .81 (for the STAMP prompt). In fact, the inter-LLM reliability for STAMP was comparable to the agreement achieved between human coders on the same tasks, confirming that STAMP produces consistent and replicable measurements. Refer to Table 1 for the inter-rater reliability between LLMs and human coders.

To enhance the robustness of our findings, we also employ bootstrap resampling (1,000 iterations) with a temperature of 0 to produce deterministic output, rather than the default temperature of 1 used in the main studies. The results show that GPT-4.1 and Gemini 2.0 Flash perform indistinguishably from each other and outperform the reliability of human experts. See Web Appendix H, Table W3.

Classification Accuracy.

STAMP also produced substantial improvements in classification accuracy, as measured by F1 scores (Table 2) over the definition-only prompt. With STAMP, both GPT-4.1 and Gemini Flash 2 achieved F1 scores above the threshold of .80 for all three studies (Study 1: Classic

Peer Review Version

Luxury Style (GPT-4: .87, Gemini Flash 2: .83), Personalized Service (GPT-4: .91, Gemini Flash 2: .90)), Study 2: Recommendation (GPT-4: .92, Gemini Flash 2: .88), and Study 3: Value Confirmation (GPT-4: .83, Gemini Flash 2: .81)). These accuracy levels met or, in some cases, exceeded the performance of human crowd coders (Study 1: F1 = .82 (Classic Luxury Style), F1 = .82 (Personalized Service), Study 2: F1 = .89 (Recommendation), and Study 3: F1 = .77 (Value Confirmation), thus, demonstrating high predictive validity of the STAMP prompt.

Furthermore, we also compare F1 scores across different prompt elements, including "definition-only," "definition + inclusion," "definition + inclusion + exclusion," and the "STAMP prompt." In all cases, the results (Table 2) reveal that the STAMP prompt achieved the highest F1 score, thereby demonstrating the relevance and importance of each of its multi-layer prompt elements. Lastly, the performance trajectories of the two LLMs across the prompt elements sometimes differed, underscoring that prompt sensitivity can be model-specific. For instance, Flash 2's performance for "Classic Luxury Style" initially dropped from an F1 of .51 to .34 when only the inclusion criteria were added, suggesting that the model may have interpreted the rules too broadly, leading to a surge in false positives. Similarly, for recommendation, GPT-4.1's performance initially dipped from .84 to .76 as it transitioned from definition-only to definition + inclusion, but later recovered to .85 after the addition of explicit exclusion criteria that provided the necessary constraints. These divergences highlight a key strength of the STAMP framework in that by designing a multi-layered structured prompt and using a dual-LLM system to identify and resolve ambiguities, STAMP offers a measurement approach that uses heterogeneous models to converge on a shared and valid understanding of the construct rather than being optimized for a single model's idiosyncrasies. For STAMP classification performance with bootstrap confidence (1,000 iterations), see Web Appendix H, Table W4.

Peer Review Version

Table 1. Inter-rater Reliability Between LLMs and Human Coders

Construct & Prompt	Krippendorff's α (GPT-4.1 and Flash 2)	Disagreement Rate (GPT-4.1 and Flash 2)	Krippendorff's α (Human Expert / Crowd)	Disagreement Rate (Human Expert / Crowd)	Krippendorff's α (Trained RA / Crowd)	Disagreement Rate (Trained RA / Crowd)
Study 1						
Classic Luxury Style – Definition-Only	.37	11.31%	N/A	N/A	N/A	N/A
Classic Luxury Style – STAMP Prompt	.77	3.70%	.80	2.80%	.69	4.00%
Personalized Service – Definition-Only	.55	9.43%	N/A	N/A	N/A	N/A
Personalized Service – STAMP Prompt	.87	2.01%	.86	2.01%	.84	2.22%
Study 2						
Recommendation – Definition-Only	.55	22.33%	N/A	N/A	N/A	N/A
Recommendation – STAMP Prompt	.69	15.00%	.71	14.00%	.47	24.83%
Study 3						
Value Confirmation – Definition-Only	.67	10.64%	N/A	N/A	N/A	N/A
Value Confirmation – STAMP Prompt	.81	2.91%	.75	3.91%	N/A	N/A

Note: Human Expert / Crowd comparison is used as the primary benchmark for human-human performance. RA denotes Research Assistant

Peer Review Version

Table 2. F1 Scores for Different Prompt Structures by Model and Construct

Studies	Study 1				Study 2		Study 3	
Prompt Structure	GPT-4.1 (Classic Luxury Style)	Flash 2 (Classic Luxury Style)	GPT-4.1 (Personalized Service)	Flash 2 (Personalized Service)	GPT-4.1 (Recomme-ndation)	Flash 2 (Recomme-ndation)	GPT-4.1 (Value Confirmation)	Flash 2 (Value Confirmation)
Definition only	.51	.51	.73	.67	.84	.86	.47	.36
Definition + Inclusion	.81	.34	.90	.73	.76	.87	.55	.49
Definition + Inclusion + Exclusion	.76	.52	.90	.83	.85	.88	.81	.78
STAMP Prompt	**.87**	**.83**	**.91**	**.90**	**.92**	**.88**	**.83**	**.81**
Human Crowd Benchmark	.82	.82	.88	.88	.89	.89	.77	.77

Peer Review Version

Construct Validity.

Beyond accuracy, the results from all three studies confirm that STAMP's classifications also adhere to theoretical construct boundaries. We use multi-trait-multi-method (MTMM) analysis to assess the convergent and discriminant validity of the constructs using GPT-4.1 measure or the state-of-the-art text model. For Study 1, the measures for the "Classic Luxury Style" and "Personalized Service" demonstrate strong construct validity. Specifically, the correlations between measurements of the same construct using two different methods (STAMP prompt vs. a definition-only prompt) were high, indicating convergent validity (Classic Luxury Style: .50, $p < .001$; Personalized Service: .77, $p < .001$). However, the correlations between Classic Luxury Style and Personalized Service were low (.06, $p = .046$), indicating discriminant validity (See Table 3). A formal Steiger's Z-test confirms that each mono-trait-hetero-method correlation was significantly greater than any hetero-trait correlation ($p < .001$), thus supporting the fact that the two constructs were distinct concepts.

Table 3. Multi-trait-Multi-Method Matrix for Study 1

No.	Construct Measurement	1	2	3	4
1	Classic Luxury Style – STAMP	1.00			
2	Classic Luxury Style – Definition-only	**.50**	1.00		
3	Personalized Service – STAMP	.06	.08	1.00	
4	Personalized Service – Definition-only	.08	.08	**.77**	1.00

Note: Mono-trait-hetero-method correlations (bolded) indicate convergent validity. Hetero-trait correlations indicate discriminant validity

To assess the construct validity of the recommendation construct (Study 2), we performed a similar analysis as we did for Study 1. The STAMP-based measure of

Peer Review Version

recommendation showed a strong correlation with the definition-only prompt (.73, $p < .001$). Furthermore, it demonstrated discriminant validity from a related but conceptually distinct construct, sentiment (See Table 4). The correlation between recommendation and sentiment was moderate (.62, $p < .001$), and this value was statistically lower than the convergent validity correlation ($Z = 4.15$, $p < .001$). The results thus confirm that STAMP successfully captured recommendations in the review text, beyond positive or negative sentiment.

Table 4. Multi-trait-Multi-Method Matrix for Recommendation vs. Sentiment (Study 2)

No	Construct Measurement	1	2	3
1	Recommendation – STAMP	1.00		
2	Recommendation – Definition-only	**.73**	1.00	
3	Sentiment - Definition-only	.62	.53	1.00

Note: Mono-trait-hetero-method correlations (bolded) indicate convergent validity. Hetero-trait correlations indicate discriminant validity.

To assess the construct validity of the value confirmation construct (Study 3), we computed the correlation between the definition-only and the STAMP-based measure. As reported in Table 5, this correlation was moderate (.48, $p <.001$). To establish discriminant validity, we also labeled our text data for a related but conceptually distinct tactic called "concession." Concession is defined as a discretionary unadvertised discount or service offered or counter-offered by a sales agent in response to a customer's bargaining request to motivate purchase. The correlation between Value confirmation and Concession tactic was low (-.09, $p = 0.007$), indicating discriminant validity. A formal Steiger's Z-test confirmed that each mono-trait-hetero-method correlation was significantly greater than any hetero-trait correlation ($Z = 4.23$, $p < .001$), providing support for the fact that the two constructs (value confirmation and concession) were measured as distinct concepts.

Peer Review Version

Peer Review Version

Table 5. Multi-trait-Multi-Method Matrix for Study 1

No.	Construct Measurement	1	2	3	4
1	Value Confirmation – STAMP	1.00			
2	Value Confirmation – Definition-only	**.48**	1.00		
3	Concession – STAMP	-.09	-.12	1.00	
4	Concession – Definition-only	-.10	-.14	**.76**	1.00

Note: Mono-trait-hetero-method correlations (bolded) indicate convergent validity. Hetero-trait correlations indicate discriminant validity

Robustness Checks.

Although we ensured that the dual-LLM classifier uses LLMs that have similar capabilities (e.g., GPT-4.1: MMLU = 90.2%, and Gemini Flash 2: MMLU-Pro = 77%) and belong to diverse development families, we also assessed whether the classifications we obtained are robust to (a) model choice within the same family (GPT-4.1 and GPT-4o), (b) model choice across major families (Claude Sonnet 4 and Gemini 2.5 Flash), and (c) the newest model generation OpenAI GPT-5 (Medium) against human generated labels. For all comparisons, we focused on the "personalized service" (Study 1) and report these results in Table 6. Results indicate that within-family alignment is high. In other words, GPT-4.1 and GPT-4o yield near-identical classifications (Krippendorff α = .85). Further, STAMP drives cross-family generalization. Specifically, Claude Sonnet 4 and Gemini 2.5 Flash achieved high agreement with STAMP (F1 = 94%, α = .93) but weak agreement with the definition-only prompt (F1 = 67%, α = .62). Lastly, the latest model (GPT5) performed similarly to GPT4.1 vs. humans. Together, these results demonstrate that STAMP is essential for high cross-family reliability and that model provider matters less once prompting is standardized.

Peer Review Version

Collectively, the results across all our studies provide evidence that STAMP produces measures that are not only accurate and reliable but also theoretically sound (i.e., respecting the conceptual boundaries between different constructs), and consistent across LLM model families.

Table 6. Cross-model Agreement on the Personalization Tactic

Comparison	Prompt	F1	Disagreement	Krippendorff's α
GPT-4.1 vs GPT-4o	STAMP	.86	2.40%	.85
GPT-4.1 vs GPT-4o	Definition	.84	2.30%	.83
Claude Sonnet 4 vs Gemini 2.5 Flash	STAMP	.93	1.21%	.92
Claude Sonnet 4 vs Gemini 2.5 Flash	Definition	.67	9.49%	.62
Researcher vs GPT-5 (Medium)	STAMP	.92	1.21%	.91
Researcher vs GPT-4.1	STAMP	.92	1.30%	.91

Discussion

STAMP advances how LLMs can be used for text classification by explicitly aligning LLM capabilities with the requirements of rigorous measurement. Rather than treating prompting as an opaque, ad-hoc interface, STAMP recasts it as a structured theory-embedded procedure and converts inter-LLM disagreement from noise into a source of diagnostic information. As LLMs become increasingly powerful, the temptation is to use them as black-box annotators or even to replace human analysts entirely. STAMP offers an alternative path. A human–LLM collaborative system in which theory guides the machine and the machine amplifies the theorist. This hybrid approach will yield meaningful and trustworthy analyses as we confront the vast and complex textual data of the digital age. By marrying computational scalability with the validity demands of social science, STAMP represents a step toward robust hybrid workflows that have

Peer Review Version

theoretical, methodological, and practical implications for marketing research, as well as associated fields grappling with making sense of big text data in a principled way.

Theoretical and Methodological Contribution

Bridging Validity and Scale.

First and foremost, STAMP demonstrates a viable path to reconcile the trade-off between interpretive depth and analytic scale. By embedding Churchill's (1979) measurement paradigm within an LLM prompt, STAMP shows that the prompt sensitivity problem can be mitigated and provides a blueprint for developing and validating theoretically grounded automated measures (MacKenzie et al. 2011). Methodologically, STAMP offers a standardized, transparent, and replicable protocol for text classification. This represents a significant step forward from the idiosyncratic, "one-off" prompting techniques that have characterized early LLM applications in the social sciences (Schulhoff et al. 2025) and ensures that the classifications produced by LLMs truly represent the intended latent constructs, providing greater confidence in the insights derived from textual data. Vague zero-shot queries can be replaced with structured, theory-based instruments, and LLM-based coding can be scaled to thousands of texts without compromising on construct validity. STAMP functions as glass box in which definition, criteria, and exemplars are explicit and inspectable. The fact that LLM coders can reach super-human reliability while adhering to a human-crafted codebook suggests that researchers no longer have to trade off quality for quantity in automated text analysis. The multi-layer structured prompt, dual-LLM reliability checks, and structured adjudication process create a framework that can be adopted and adapted by other researchers for text-based constructs in varied contexts. This standardization fosters a more cumulative scientific approach to text analytics, where construct measures can be compared across studies with confidence in their methodological underpinnings.

Peer Review Version

Advancing Construct Validity in LLM Applications.

STAMP reframes prompt design as a construct validity exercise rather than a purely predictive one. Much existing work with LLMs in marketing and social science emphasizes accuracy relative to human labels or downstream predictive performance, often treating the label set as given and forgoing a formal validation framework. Studies such as Gilardi et al. (2023) and Bojić et al. (2025) demonstrate that LLMs can produce inexpensive, rapid labels with accuracy comparable to that of experts; however, they typically assess success in terms of agreement rather than adherence to theoretical definitions. STAMP addresses this gap by embedding construct definition and discriminant criteria directly into the prompt and by treating reliability as a precursor to validity. Once high inter-LLM agreement is achieved, STAMP subjects the resulting labels to validity checks (convergent and discriminant validity via multi-trait multi-method matrices). This brings LLM-based coding closer to the standards applied to new psychometric scales. In doing so, it positions LLMs not merely as annotators but as components of a theory-driven measurement system.

Disagreement as Theoretical Discovery.

A novel contribution of STAMP is its treatment of disagreements as a source of theoretical insight rather than noise. Traditional intercoder reliability checks typically treat low agreement as a failure to be corrected, but the disagreements themselves are rarely analyzed. STAMP instead logs and analyzes disagreements. Consistent with Jayaweera and Dorr (2025) and the CrowdTruth framework (Aroyo and Welty 2015), repeated patterns of disagreement are used to identify ambiguous terms, underspecified construct boundaries, or instances of label pluralism. The rationales generated by LLMs provide qualitative insights at scale, enabling

Peer Review Version

researchers to see why cases are contested and to refine definitions or coding rules accordingly. In this way, reliability assessments become a driving force for theoretical refinement.

Hybrid AI–Human Workflows in Research.

STAMP also formalizes a hybrid LLM-human workflow. In this setup, LLMs handle the bulk of routine coding while human researchers concentrate on the cases that matter the most for theory (ambiguous, contested, and sensitive). This division of labor aligns with evidence from hybrid decision systems (e.g., Chakraborty et al. 2025), where combining AI with human oversight outperforms AI-only approaches by 67%. STAMP explicitly seeks guidance from a human expert for defining constructs, refining prompts, and adjudicating disagreement cases, thereby addressing concerns that unchecked AI could propagate bias or overconfidence errors at scale (Carlson and Burbano 2025; Liu et al. 2023).

Finally, STAMP enhances explainability and transparency. Each classification is accompanied by a model-generated rationale tied to the codebook definition and criteria. Disagreement cases include contrasting rationales and a synthesized summary of the arguments. This goes beyond post hoc feature-based explanations (e.g., SHAP or LIME) by requiring the model to justify its decisions in terms of the construct itself. The result is an audit trail that allows reviewers to inspect how constructs were operationalized, why particular texts received particular labels, and how disagreements were resolved. Because the prompt and rationale structure can be applied across different LLMs, STAMP also encourages cross-model robustness checks, helping ensure that findings are not artifacts of a single proprietary model. Collectively, these features support more responsible, transparent, and theoretically grounded uses of AI in marketing and related fields.

Peer Review Version

Practical Implications

STAMP formalizes a hybrid division of labor in which LLMs handle scalable, routine coding while humans focus on theory building, adjudicating edge cases, and refining constructs. This design aligns with emerging AI–human hybrid frameworks, which demonstrate that combining LLM-based coding with expert judgment enhances both predictive performance and interpretability in marketing contexts (Arora et al. 2024; Chakraborty et al. 2025). At the same time, recent work on LLM-assisted annotation stresses the continued need for human-written codebooks and human validation to guard against biased or overconfident outputs (Liu et al. 2023; Villarroel Ordenes et al. 2025). Within STAMP, humans thus act as judges of judgment, interpreting disagreement patterns, updating prompts, and adjudicating ambiguous cases, while LLMs provide consistent, large-scale coding that would be infeasible for humans alone.

Historically, the production of human-labeled data served as a bottleneck in machine learning applications, requiring expert services or large-scale crowdsourcing to train specialized models. STAMP inverts this workflow. Researchers can use STAMP to generate high-quality, theory-driven labels, which can subsequently be used to fine-tune smaller, open-source models to achieve flagship-level performance (Bermejo et al. 2025). This alters the traditional sequence from human-first data generation to an AI-first approach with targeted human validation. This result is an accelerated research timeline and broader access to sophisticated text analysis by reducing reliance on resource-intensive manual coding (Gilardi et al. 2023).

Responding to calls for greater practical impact and accessibility in marketing scholarship, methodological contributions must be usable by non-technical audiences (American Marketing Association, n.d.). To that end, STAMP is implemented as an interactive, solution-focused web application in which methodological innovations are delivered as dynamic

Peer Review Version

computational tools that facilitate adoption and usage in practice (American Marketing Association, n.d.; Blanchard et al. 2025). The app addresses multiple technical challenges, including API calls, rate limits, timeouts, parallel processing, XML/JSON parsing, and integrating heterogeneous LLM input/output formats. By lowering the technical barrier to entry, the STAMP application enables brand managers, market researchers, and policy analysts to deploy rigorous, LLM-based text classification without specialized programming expertise.

For practitioners and academics alike, the STAMP app provides a scalable alternative to costly manual coding and less flexible dictionary-based methods (Bermejo et al. 2025; Humphreys and Wang 2018). Using current official API prices as a reference, a typical STAMP round on 1,000 rows (approximately 1.6 million tokens) costs between $0.33 and $0.68 with Gemini (Batch) and between $1.35 and $2.75 with GPT (Batch). A typical optimization cycle of two iterations costs between $1.00 and $4.10 per 1,000 items. Even scaling to 70,000 rows yields estimated costs of only $70.60 for Gemini and $287 for GPT. These figures are orders of magnitude lower than the approximately $600 per 1,000 items typically incurred with crowd coding (e.g., 4 coders × 8 hours × $15/hour), consistent with findings that LLM annotation can be vastly cheaper than crowdsourcing while maintaining high quality (Gilardi et al. 2023). In practical terms, STAMP enables organizations to analyze customer feedback and social media conversations with greater confidence, speed, and scale, while preserving theoretical rigor.

Limitations and Future Research

While this research establishes the validity and utility of STAMP, it also presents limitations that suggest avenues for future inquiry. The validation relies on two proprietary state-of-the-art LLMs and on English-language text. Given the rapid evolution of LLMs, new models are constantly emerging. Future research could examine STAMP's performance in comparison to

Peer Review Version

other models, particularly open-source alternatives, which may offer cost and accessibility benefits. Furthermore, extending the validation to other languages is crucial to assess STAMP's cross-lingual generalizability. Although STAMP is efficient as compared to manual coding, the use of state-of-the-art LLMs via APIs incurs additional computational costs. For extremely large-scale applications involving millions of documents, these costs could become substantial. Future work could explore the efficacy of smaller, specialized, or fine-tuned LLMs within the STAMP framework to optimize the cost-performance trade-off without sacrificing measurement rigor.

Peer Review Version

References

Abdurahman, Shadi, Mohammad Atari, Fatemeh Karimi-Malekabadi, Melissa J. Xue, Jesse Trager, Peter S. Park, Peter S. Golazizian, Ashwini Omrani, and Morteza Dehghani (2024), "Perils and Opportunities in Using Large Language Models in Psychological Research," *PNAS Nexus*, 3 (7), pgae245. https://doi.org/10.1093/pnasnexus/pgae245.

American Marketing Association (n.d.), "Journal of Marketing Call for Papers: Marketing Impact with Research-Driven Apps," [Special issue call for papers].

Anthropic (2025), "Use XML Tags to Structure Your Prompts," Anthropic Documentation.

Arora, Neeraj, Ishita Chakraborty, and Yohei Nishimura (2024), "AI–Human Hybrids for Marketing Research: Leveraging LLMs as Collaborators," *Journal of Marketing*, 89 (3), Advance online publication. https://doi.org/10.1177/00222429241276529.

Aroyo, Lora and Chris Welty (2015), "Truth Is a Lie: CrowdTruth and the Seven Myths of Human Annotation," *AI Magazine*, 36 (1), 15–24.

Artstein, Ron and Massimo Poesio (2008), "Inter-Coder Agreement for Computational Linguistics," *Computational Linguistics*, 34 (4), 555–96. https://doi.org/10.1162/coli.07-034-R2.

Balakrishnan, Maya, Karen J. Ferreira, and Jordan Tong (2025), "Human–Algorithm Collaboration with Private Information: Naïve Advice-Weighting Behavior and Mitigation," *Management Science*, Advance online publication. https://doi.org/10.1287/mnsc.2022.03850.

Berger, Jonah, Ashlee Humphreys, Stephan Ludwig, Wendy W. Moe, Oded Netzer, and David A. Schweidel (2020), "Uniting the Tribes: Using Text for Marketing Insight," *Journal of Marketing*, 84 (1), 1–25. https://doi.org/10.1177/0022242919873106.

Bermejo, Victor J., Alberto Gago, Ramiro H. Gálvez, and Nimrod Harari (2025), "LLMs Outperform Outsourced Human Coders on Complex Textual Analysis," *Scientific Reports*, 15, 40122. https://doi.org/10.1038/s41598-025-23798-y.

Blanchard, Simon J., Nicole Duani, Andrew M. Garvey, Oded Netzer, and Thomas T. Oh (2025), "New Tools, New Rules: A Practical Guide to Effective and Responsible Generative AI Use for Surveys and Experiments in Research," *Journal of Marketing*, 89 (6), 119–39. https://doi.org/10.1177/00222429251349882.

Bojić, Ljubiša, Olena Zagovora, Agne Zelenkauskaite, Viktor Vuković, Milica Čabarkapa, Sanja Veseljević Jerković, and Aleksandra Jovančević (2025), "Comparing Large Language Models and Human Annotators in Latent Content Analysis of Sentiment, Political Leaning, Emotional Intensity and Sarcasm," *Scientific Reports*, 15 (1), 11477. https://doi.org/10.1038/s41598-025-96508-3.

Peer Review Version

Carlson, Nora A. and Vanessa Burbano (2025), "The Use of LLMs to Annotate Data in Management Research: Foundational Guidelines and Warnings," *Strategic Management Journal*, 46 (forthcoming). https://doi.org/10.1002/smj.70023.

Chakraborty, Ishita, Kyu Chiong, Harry Dover, and K. Sudhir (2025), "Can AI and AI-Hybrids Detect Persuasion Skills? Salesforce Hiring with Conversational Video Interviews," *Marketing Science*, 44 (1), 30–53. https://doi.org/10.1287/mksc.2023.0149.

Chakraborty, Ishita, Minki Kim, and K. Sudhir (2022), "Attribute Sentiment Scoring with Online Text Reviews: Accounting for Language Structure and Missing Attributes," *Journal of Marketing Research*, 59 (3), 600–22.

Churchill, Gilbert A. (1979), "A Paradigm for Developing Better Measures of Marketing Constructs," *Journal of Marketing Research*, 16 (1), 64–73.

Corbin, Juliet and Anselm Strauss (1990), "Grounded Theory Research: Procedures, Canons, and Evaluative Criteria," *Qualitative Sociology*, 13 (1), 3–21.

Gilardi, Fabrizio, Meysam Alizadeh, and Maël Kubli (2023), "ChatGPT Outperforms Crowd Workers for Text-Annotation Tasks," *Proceedings of the National Academy of Sciences*, 120 (30), e2305016120. https://doi.org/10.1073/pnas.2305016120.

Goli, Amin and Abhi Singh (2024), "Frontiers: Can Large Language Models Capture Human Preferences?" *Marketing Science*, 43 (4), 709–22. https://doi.org/10.1287/mksc.2023.0306.

Google Cloud (2025), "Structure Prompts," Google Cloud Documentation.

Hendrycks, Dan, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt (2021), "Measuring Massive Multitask Language Understanding," (No. arXiv:2009.03300), arXiv. https://doi.org/10.48550/arXiv.2009.03300.

Hohenwalde, Carolin, Maike Leidecker-Sandmann, Nele Promies, and Markus Lehmkuhl (2025), "ChatGPT's Potential for Quantitative Content Analysis: Categorizing Actors in German News Articles," *Journal of Science Communication*, 24 (2), A01. https://doi.org/10.22323/2.24020201.

Humphreys, Ashlee and Rebecca Jen-Hui Wang (2018), "Automated Text Analysis for Consumer Research," *Journal of Consumer Research*, 44 (6), 1274–306. https://doi.org/10.1093/jcr/ucx104.

Jayaweera, Chiranthi and Bonnie J. Dorr (2025), "From Disagreement to Understanding: The Case for Ambiguity Detection in NLI," arXiv Preprint arXiv:2507.15114. https://doi.org/10.48550/arXiv.2507.15114.

Krippendorff, Klaus (2018), *Content Analysis: An Introduction to Its Methodology*, 3rd ed. Thousand Oaks, CA: Sage Publications.

Peer Review Version

Laurito, William, Ben Davis, Paul Grietzer, Tomáš Gavenčiak, Adam Böhm, and Jan Kulveit (2025), "AI–AI Bias: Large Language Models Favor Communications Generated by Large Language Models," *Proceedings of the National Academy of Sciences*, 122 (31), e2415697122. https://doi.org/10.1073/pnas.2415697122.

Lin, Zheng (forthcoming), "From Prompts to Constructs: A Dual-Validity Framework for LLM Research in Psychology," *Annual Review of Psychology*.

Liu, Yang (2023), "The Importance of Human-Labeled Data in the Era of LLMs," in *Proceedings of the 32nd International Joint Conference on Artificial Intelligence (IJCAI 2023)*, (Article 802). https://doi.org/10.24963/ijcai.2023/802.

Lundberg, Scott M. and Su-In Lee (2017), "A Unified Approach to Interpreting Model Predictions," *Advances in Neural Information Processing Systems*, 30.

MacKenzie, Scott B., Philip M. Podsakoff, and Nathan P. Podsakoff (2011), "Construct Measurement and Validation Procedures in MIS and Behavioral Research: Integrating New and Existing Techniques," *MIS Quarterly*, 35 (2), 293–334. https://doi.org/10.2307/23044045.

Mu, Y., B.P. Wu, W. Thorne, et al. (2024), "Navigating Prompt Complexity for Zero-Shot Classification: A Study of LLMs in Computational Social Science," in *Proceedings of the LREC–COLING 2024 Conference*, 12074–86.

Rocklage, Matthew D., Derek D. Rucker, and Loran F. Nordgren (2018), "The Evaluative Lexicon 2.0: The Measurement of Emotionality, Extremity, and Valence in Language," *Behavior Research Methods*, 50 (4), 1327–44. https://doi.org/10.3758/s13428-017-0975-6.

Schulhoff, Seth, M. Ilie, N. Balepur, K. Kahadze, A. Liu, C. Si, Y. Li, et al. (2025), "The Prompt Report: A Systematic Survey of Prompt Engineering Techniques," arXiv Preprint arXiv:2406.06608. https://doi.org/10.48550/arXiv.2406.06608.

Sirianni, Nancy J., Mary Jo Bitner, Stephen W. Brown, and Naomi Mandel (2013), "Branded Service Encounters: Strategically Aligning Employee Behavior with the Brand Positioning," *Journal of Marketing*, 77 (6), 108–23.

Snow, Rion, Brendan O'Connor, Daniel Jurafsky, and Andrew Y. Ng (2008), "Cheap and Fast – But Is It Good? Evaluating Non-expert Annotations for Natural Language Tasks," in *Proceedings of the 2008 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 254–63.

Tausczik, Yla R. and James W. Pennebaker (2010), "The Psychological Meaning of Words: LIWC and Computerized Text Analysis Methods," *Journal of Language and Social Psychology*, 29 (1), 24–54. https://doi.org/10.1177/0261927X09351676.

Peer Review Version

Turpin, Miles, Julian Michael, Ethan Perez, and Samuel Bowman (2023), "Language Models Don't Always Say What They Think: Unfaithful Explanations in Chain-of-Thought Prompting," in *Advances in Neural Information Processing Systems*, 36 (NeurIPS 2023).

Uma, Alexandra, Danae Almanea, and Massimo Poesio (2022), "Scaling and Disagreements: Bias, Noise, and Ambiguity," *Frontiers in Artificial Intelligence*, 5, 818451. https://doi.org/10.3389/frai.2022.818451.

Villarroel Ordenes, Francisco, Grant Packard, Jochen Hartmann, and Davide Proserpio (2025), "Using Traditional Text Analysis and Large Language Models in Service Failure and Recovery," *Journal of Service Research*, 28 (3), 382–8. https://doi.org/10.1177/10946705241307678.

Wang, L., X. Chen, X. Deng, H. Wen, M. You, W. Liu, Q. Li, and J. Li (2024), "Prompt Engineering in Consistency and Reliability with the Evidence-Based Guideline for LLMs," *npj Digital Medicine*, 7 (1), 41. https://doi.org/10.1038/s41746-024-01029-4.

Zheng, Lianmin, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, et al. (2023), "Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena," (No. arXiv:2306.05685), arXiv. https://doi.org/10.48550/arXiv.2306.05685.

Ziems, Caleb, William Held, Omar Shaikh, Jiaao Chen, Zhehao Zhang, and Diyi Yang (2024), "Can Large Language Models Transform Computational Social Science?" *Computational Linguistics*, 50 (1), 237–91. https://doi.org/10.1162/coli_a_00502.

Peer Review Version

STAMP: A Standardized Protocol and Web Application for LLM-based Text Classification

Web Appendix

Table of Contents

Web Appendix A: Comparing Extant Text Classification Methods (Dictionary, Supervised ML) with STAMP.	1
Web Appendix B: STAMP Web Application User Guide	3
Web Appendix C: Review of Prior Studies That Used Multiple Validity Checks	6
Web Appendix D: Comparison of Single LLM-based Classification with STAMP	7
Web Appendix E: LLM Model Selection Choices and Cross-Model Generalizability	9
Web Appendix F: STAMP Workflow (Prompts, Disagreements, Prompt Iterations) for Construct Classification	11
Web Appendix G: STAMP Workflow from Definition-Only to Full STAMP Prompt (Personalized Service Example)	16
Web Appendix H: Robustness of Results Across Models.	20
References	22

Disclosure: The authors have supplied these materials to aid in understanding their paper.

Peer Review Version

Web Appendix A: Comparing Extant Text Classification Methods (Dictionary, Supervised ML) with STAMP

Dimension	**Dictionary (LIWC, DICTION)**	**Supervised ML (e.g., SVM, Random Forest, CNN/RNN, fine-tuned BERT)**	**LLM - generative AI only (e.g., GPT, Claude, Gemini)**
Mechanism used	Static lexicons/keyword lists	Trained classifiers on labeled data	Multi-layer structured prompt, dual-LLM agreement + HITL adjudication
Data/label needs	None (but lexicon building is domain-specific)	High (large hand-labeled datasets needed for each construct/domain)	Low (zero-shot or few-shot; minimal exemplars needed) (Chae and Davidson 2025)
Context/polysemy handling	Weak (cannot disambiguate polysemous words or interpret context) (Humphreys and Wang 2018)	Dependent on the training domain and feature engineering	Strong (context-aware LLMs with structured prompts defining construct, criteria, and examples)
Transparency	Transparent (explicit word lists)	Opaque (black box with no direct explanation for predictions) (Villarroel Ordenes et al. 2025)	Transparent (chain-of-thought reasoning and structured outputs enable auditability)(Wei et al. 2022)
Validity	Face validity present; limited construct validity (Short et al. 2010)	Accuracy-focused; construct validity is rarely evaluated (Humphreys and Wang 2018)	Built-in content, convergent, and discriminant validity checks through multi-method comparisons

Peer Review Version

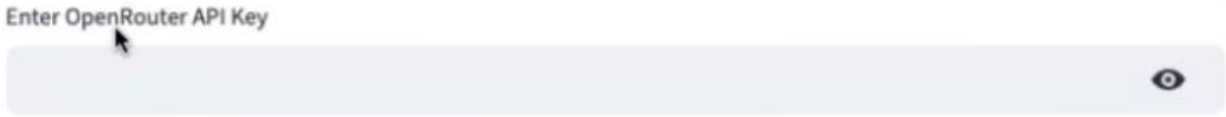
Dimension	Dictionary: LIWC, DICTION)	Supervised ML (e.g., SVM, Random Forest, CNN/RNN, fine-tuned BERT)	LLM - generative AI only (e.g., GPT, Claude, Gemini)
Reliability	Inconsistent (brittle across domains; struggles with context) (Hartmann et al. 2023)	Cross-validation only; no inter-rater reliability analogy	High reliability via multiple independent LLM "raters" with agreement thresholds; disagreements resolved by human experts
Replicability/ standardization	Replicable within the domain; brittle and domain-specific across contexts	Reproducible with code/data; not portable without retraining for new constructs	Highly portable and replicable standardized protocol across constructs and contexts
Scalability	High (once the lexicon is built)	Moderate (requires new labeled datasets for each application)	High (leverage pre-trained LLMs; no task-specific training required)
Key limitations	Cannot interpret context; misclassifies ambiguous cases	Requires extensive labeling; lacks transparency and validity checks; prone to hidden biases	Dependent on prompt quality, may struggle with entirely new theoretical constructs

Peer Review Version

Web Appendix B: STAMP Web Application User Guide

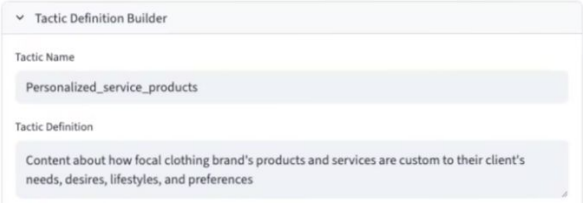
We outline the key steps a user needs to follow in the web application. We also provide a 5-minute video tutorial (https://figshare.com/s/7442f7dc932ae41f4126?file=59126732).

1. Setup: The user first obtains API access to LLMs via a platform like OpenRouter (openrouter.ai). After securing the necessary API key (for example, an OpenAI GPT-4.1 key and access to another model), the user launches the web app in a browser (no local installation required). The user enters their API credentials into the interface to enable the LLMs.



2. Data Upload: The user prepares a CSV file containing the text data to be analyzed, where each text instance (e.g., a sentence, review, or social media post) is in a separate row (with any identifier or metadata fields as needed). The app provides prompts to ensure the data is in the correct format, and the user uploads the file via the interface.

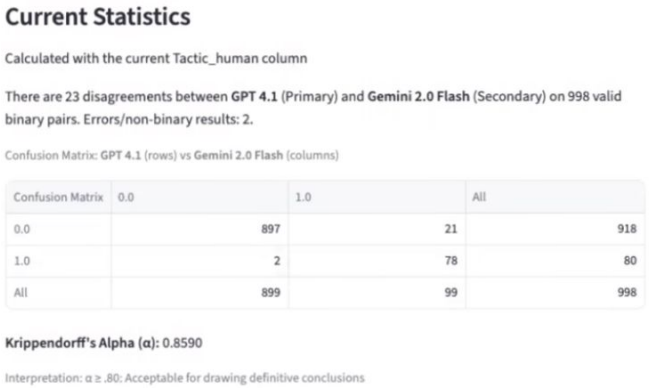
3. Prompt Construction: The app then guides the user in constructing the STAMP prompt for the chosen construct. The interface presents dedicated fields or templates for each component of the structured prompt: the construct definition, inclusion criteria, exclusion criteria, and exemplar examples. The user fills in each section (guidance and pre-filled examples are provided for reference). The app automatically formats these inputs into the structured prompt syntax (with the appropriate tags) required by the STAMP protocol. This guided prompt-builder ensures that even users unfamiliar with prompt engineering can create a robust, properly formatted prompt for their construct of interest.



Review Version

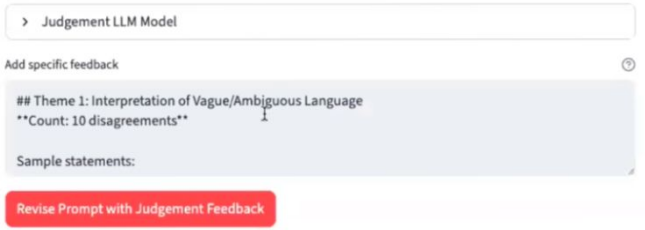
4. Run STAMP Classification: Once the prompt is configured, the user initiates the classification run. For each piece of text data, the app sends the structured prompt along with the text instance to two different LLMs (by default, GPT-4.1 and a second top-performing model such as Gemini Flash 2 from Google DeepMind) in parallel. The models' classification outputs—including their step-by-step reasoning (if available)—are returned to the app. The platform automatically computes inter-model agreement statistics (Krippendorff's α and percent agreement) in real time as the data are being coded, giving the user immediate feedback on reliability.

5. Iterate Prompt if Needed: If the initial inter-LLM agreement falls below the desired threshold (α < .67, indicating insufficient reliability), the app facilitates an iterative refinement process. It highlights the specific text instances where the two LLMs disagreed and uses a third LLM as an adjudicator to summarize common reasons for these disagreements (e.g., ambiguous wording, edge-case scenarios, misinterpreted criteria). Based on this feedback, the user can adjust the prompt by clarifying the construct definition, adding or refining inclusion/exclusion rules, or providing additional exemplars to address confusion. The user then re-runs the classification on the dataset with the revised prompt. This iteration loop can continue until agreement between the models meets or exceeds the threshold (Krippendorff's α > .67), indicating a reliably coded dataset (Krippendorff 2018).

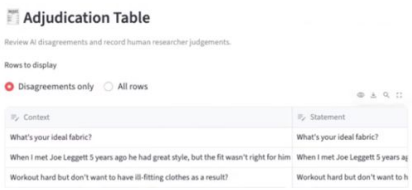


Confusion Matrix	0.0	1.0	All
0.0	897	21	918
1.0	2	78	80
All	899	99	998

Review Version



6. Human Adjudication: Once inter-LLM reliability is achieved, any remaining individual instances of disagreement between the two LLMs are flagged for human review. For these cases, the app presents the text alongside the two models' outputs and rationales. A human coder (the user or a domain expert) can then examine the evidence and make the final determination for each disputed item. This ensures that the final labeled dataset resolves any subtle ambiguities that even the LLMs could not agree upon.



7. Results and Export: After classification (and human adjudication of any flagged cases), the app compiles the final set of labels for all texts. It also generates a summary report detailing the models' performance, including final agreement metrics and classification accuracy (if ground-truth labels or proxy measures like star ratings were available for comparison). The user can download the labeled dataset, the model-generated reasoning traces for each classification, and the summary report for documentation or further analysis.

By streamlining these steps, the web application lowers the barrier to implementing STAMP in both research and industry settings. Users can achieve reliable, valid construct classifications on their text data in a transparent and efficient manner, without requiring programming expertise. In essence, the tool operationalizes our framework and makes it accessible: marketing scholars can use it to analyze content with improved rigor and replicability, and practitioners (e.g., brand managers or analysts) can confidently deploy LLM-based content analysis knowing that the classifications are grounded in a validated protocol. This accessibility enables the translation of our methodological contributions into real-world impact.

Peer Review Version

Web Appendix C: Prior Studies That Used Multiple Validity Checks

Research	Text Classification Method	Validity Type(s) Assessed
(Netzer et al. 2012)	Machine Learning- Unsupervised (topic modeling for Market Structure)	Convergent validity: ✓; Predictive validity: ✓
(Short et al. 2010)	Dictionary-based (custom word lists for Entrepreneurial Orientation)	Content validity: ✓; Discriminant validity: ✓; Predictive validity: ✓
(Ludwig et al. 2013)	Dictionary-based (LIWC for emotional tone + linguistic style matching algorithms)	Content validity: ✓; Predictive validity: ✓
(Timoshenko and Hauser 2019)	Machine Learning-Supervised (CNN classifier to filter & cluster online review sentences for customer needs)	Concurrent validity: ✓; Predictive validity: ✓
(Yeomans 2021)	Compare multiple methods (dictionary vs. domain-specific models for concreteness)	Construct validity: ✓; Predictive validity: ✓
(Martin and Kushwaha 2024)	Machine Learning (Top Management Team language (marketing vs. financial goals)	Construct validity: ✓; External validity: ✓
This Research	***STAMP Protocol***	***Content validity: ✓; Construct validity (convergent and discriminant): ✓; Predictive validity: ✓***

Peer Review Version

Web Appendix D: Comparison of Single LLM-based Classification with STAMP

Study	Data	LLM's Used; Prompt(s)	New Construct (Y/N); Construct Studied	Human Baseline; Gold Standard	Validity Checks	Findings and Limitations in comparison to STAMP
Bermejo et al. (2025)	210 Spanish news articles	GPT and Claude; Zero-shot	No; Political criticism: Entity counts (Name-entity-recognition)	Students as human coders; authors (Gold Standard)	No (only focused on accuracy vs expert labels)	LLMs outperform human coders; however, no disagreement analysis is reported
Chakraborty et al. (2025)	Video interviews for hiring a salesperson	GPT; persona prompt	No; Persuasion Skills (e.g., active listening, adaptability)	Not available	Yes (only LLM scores are validated against expert labels)	AI-human improves workforce quality; assumes GPT-generated data is valid
Gilardi et al. (2023)	6,183 tweets and news articles	GPT; Zero-shot	No; Relevance, Stance, Frames	MTurk crowd workers; Trained RAs (Gold Standard)	No (evaluates accuracy and agreement against Trained RAs)	LLMs outperform crowd workers; however, no disagreement analysis is reported
Carlson and Burbano (2025)	Crowdfunding projects from Kickstarter	GPT and Claude; Chain of Thought, Zero-shot	No; Product sustainability	Human-labeled dataset previously used for ML validation	Yes (emphasis on result validity)	LLM-based annotation matches or exceeds supervised classifier in accuracy; Solves prompt sensitivity statistically, not theoretically.

Peer Review Version

Study	Data	LLM's Used; Prompt(s)	New Construct (Y/N); Construct Studied	Human Baseline; Gold Standard	Validity Checks	Findings and Limitations in comparison to STAMP
Bojić et al. (2025)	100 curated documents	GPT, Gemini, Llama, Mixtral; Zero-shot	No; Sentiment, Political Leaning, Emotion, Sarcasm	33 trained annotators (Academic experts)	Not explicit (focused on reliability metrics)	LLMs match/exceed humans' coding on sentiment/politics, but struggle with sarcasm; no solution has been identified for sarcasm.
Törnberg (2025)	500 tweets from 11 countries	GPT; Zero-shot	No; Political affiliation	Expert coders; Crowd workers; Supervised ML (BERT, Naive Bayes)	No (Accuracy/F1 vs Ground Truth - Known party affiliation)	GPT outperforms experts and supervised models; no disagreement analysis
Current Study (STAMP)	*Study 1*: Brand's Instagram (1000 posts); Study 2: Amazon Reviews (600 review statements); Study 3: Dyadic sales negotiations (1000 sales turn statements)	GPT and Gemini; Claude as Judge; Chain-of-Thought, Few-Shot, Role-Based, and Measurement Structure)	Yes; *Brand dimensions* (Classic Luxury Style and Personalized Service); Influence tactics (Recommendation); Sales tactic (Value Confirmation)	1-2 Trained RA; 9-11 Crowd majority vote; 3 Expert Adjudicator (Gold Standard)	Construct Validity (MTMM matrix); Inter-LLM Reliability (Krippendorff alpha >.67); Accuracy Performance metrics (F1, precision, recall)	STAMP achieves human-level reliability/validity; the dual-LLM approach filters hallucinations; MTMM confirms construct distinctiveness.

Peer Review Version

Web Appendix E: LLM Model Selection Choices and Cross-Model Generalizability

Models vary in intelligence and cost. Expensive models excel at tasks unrelated to text classification (e.g., multi-modal intelligence). To ensure a fair comparison, we select models at comparable levels of general intelligence using standardized benchmarks. We use Massive Multitask Language Understanding (MMLU) as an anchor for broad knowledge and reasoning (Hendrycks et al. 2021). MMLU assesses knowledge across 57 academic and professional subjects and is widely used for evaluating advanced LLMs. Because text classification depends heavily on language understanding, MMLU serves as a suitable benchmark for model selection. MMLU-Pro is the successor to MMLU and is hence used to select models in this research. Our model recommendations are based on performance (similarity to human expert—crowd alpha, F1—crowd) and cost.

We find that price does not predict performance on binary classification tasks. For example, low-cost models GPT-4.1-mini and Claude 3.5. Haiku costs much more than Google Gemini 2 Flash but significantly underperforms. Moreover, higher-intelligence and more expensive flagship models such as Claude 4.5 don't outperform GPT-4.1.

Table W1. Model Language Test and STAMP F1 Results

Model	MMLU-Pro	Avg F1 across three tactics	Precision and Recall Balance	Cost Tier	Recommendation
Claude 4.5 Sonnet	86%	0.860	Good	Expensive	Good alternative, but ceiling effect
GPT-4.1	81%	0.886	Excellent	Mid-range	Optimal flagship ✓
Gemini 2.5 Flash	81%	0.836	Good	Low-cost	Alternative to 2.0 Flash
Gemini 2.0 Flash	78%	0.863	Good	Low-cost	Optimal budget ✓
GPT-4.1-mini	78%	0.768	Poor (very low recall)	Low-cost	Avoid ✗
Claude 3.5 Haiku	63%	0.835	Poor (very low precision)	Low-cost	Avoid ✗

Peer Review Version

Table W2. Model Cost Comparison

Model	Input Cost	Output Cost	Total (1M/1M)	Relative to GPT-4.1
Gemini 2.0 Flash	$0.10	$0.40	$0.50	Baseline (20× cheaper)
Gemini 2.5 Flash	$0.15	$0.60	$0.75	15× cheaper (but 50% more than 2.0)
GPT-4.1-mini	$0.40	$1.60	$2.00	5× cheaper (but poor performance)
Claude 3.5 Haiku	$0.80	$4.00	$4.80	2× cheaper (but poor performance)
GPT-4.1	$2.00	$8.00	$10.00	Reference
Claude 4.5 Sonnet	$3.00	$15.00	$18.00	1.8× more expensive

Peer Review Version

Web Appendix F: STAMP Workflow (Prompts, Disagreements, Prompt Iterations) for Construct Classification

Full Structured Prompt (Study 1, Construct: Personalized Service)

This prompt targets whether a post indicates that the brand customizes products or services to individual clients' needs or preferences. It delineates multiple inclusion rules (explicit mentions of custom offerings, one-to-one service experiences, emphasis on individual fit) and exclusion rules (generic marketing features, non-personalized invites, etc.). Key portions of the prompt are shown below:

<Definition>

You are a text classifier to identify the tactic "Personalized Service": content showing that a brand customizes products or services to an individual client's needs, desires, lifestyle, or preferences.

<Definition>

You will be provided with two pieces of information: <context> and <statement>. Your task is to determine if the "statement" qualifies as the tactic based on the following criteria:

<Inclusion Criteria>

(if ANY are true; it's the tactic):
1. **Tailor-made offering** – mentions *custom, bespoke, made-to-measure, monograms* or other wording that indicates the product/service itself is created to a client's specifications.
2. **Explicitly personalized experience** – describes a one-to-one, private, or individualized shopping/delivery interaction (e.g., "private Zoom fitting," "one-on-one styling session," "house call tailoring," "personal wardrobe consultation").
3. **Emphasis on individual fit** – stresses that the offer adapts to the client's unique needs, desires, lifestyle, measurements, or preferences (e.g., "engineered for your lifestyle").
</Inclusion Criteria>

<Exclusion Criteria>

(if ANY are true; it's not the tactic):
1. **Standard retail variety** – ONLY discussion of unique fabrics, wide product ranges, or other features that do **not** involve tailoring to the *individual*.
2. **Generic invite without personalization** – ONLY broad calls such as "DM me for a consultation" or "Let's hop on a call" that lack any wording implying privacy, individuality, or customization.
3. **Non-text content only** – statement that contains only punctuation or other non-textual characters.

Peer Review Version

</Exclusion Criteria>

Please analyze the given statement and provide your output in the following JSON format:

{
"reasoning": "inclusion_criteria [list the criteria number] and exclusion_criteria [criteria number]; explanation: [Your step by step reasoning, highlighting keywords]",
"tactic_prediction": [0 or 1]
}

<examples>

<example>

Context: "Don't rent a tuxedo when you could have one made for you! Wear a personalized custom formal blazer made strictly for you and blow the crowd away!"

Statement to detect: "Don't rent a tuxedo when you could have one made for you! Wear a personalized custom formal blazer made strictly for you and blow the crowd away!"

{
"reasoning": "inclusion_criteria [1,3] and exclusion_criteria []; explanation: The statement emphasizes 'personalized custom' and 'made strictly for you', clearly indicating products tailored to individual needs.",
"tactic_prediction": 1
}
</example>

<example>

Context: "Unlike box subscriptions & online styling services, I can make house calls & offer custom clothing right to your door."

Statement to detect: "Unlike box subscriptions & online styling services, I can make house calls & offer custom clothing right to your door."

{
"reasoning": "inclusion_criteria [1,2] and exclusion_criteria []; explanation: The statement mentions 'house calls' (personalized service delivery) and 'custom clothing', indicating individualized service and products.",
"tactic_prediction": 1
}
</example>

<example>

Peer Review Version

Context: "I save my clients time by coming to them in their offices or their homes. I assess what you do, who you see, where you travel. Based on that information I make suggestions on cloth & construction that will best suit your needs."

Statement to detect: "I save my clients time by coming to them in their offices or their homes... Based on that information I make suggestions on cloth & construction that will best suit your needs."

{
"reasoning": "inclusion_criteria [2,3] and exclusion_criteria []; explanation: The statement describes personalized service delivery (house/office visits), individual assessment, and customization based on client's specific lifestyle and needs.",
"tactic_prediction": 1
}
</example>

<example>

Context: "Personalized however you want! That's my favorite part about custom & who doesn't love clothing that fits?"

Statement to detect: "Personalized however you want! That's my favorite part about custom & who doesn't love clothing that fits?"

{
"reasoning": "inclusion_criteria [1,3] and exclusion_criteria []; explanation: The statement explicitly uses 'Personalized however you want' and mentions 'custom', directly indicating individualized products and services.",
"tactic_prediction": 1
}
</example>

<example>

Context: "Browse unique one-of-a-kind fabrics along with the last pieces of our best selling fabrics to complement your wardrobe."

Statement to detect: "Browse unique one-of-a-kind fabrics along with the last pieces of our best selling fabrics to complement your wardrobe."

{
"reasoning": "inclusion_criteria [] and exclusion_criteria [1]; explanation: The statement only discusses unique fabrics and product variety without mentioning any personalization, customization, or individual tailoring aspects.",
"tactic_prediction": 0
}

Peer Review Version

</example>

<example>

Context: "Give mom the gift of luxury & the custom experience this Mother's Day!"

Statement to detect: "Give mom the gift of luxury & the custom experience this Mother's Day!"

{
"reasoning": "inclusion_criteria [1,3] and exclusion_criteria []; explanation: The statement mentions 'custom experience', indicating personalized service tailored to individual preferences.",
"tactic_prediction": 1
}
</example>

</examples>

Model Outputs and Agreement: Using this prompt, the same two LLMs (GPT-4.1 and Gemini Flash 2) were coded to generate personalized-service content from the luxury post data. Inter-model agreement for this construct reached Krippendorff's α = .82, reflecting high consistency (notably higher than the zero-shot definition-only baseline for this construct, which yielded α = .55 in initial trials). Only around 5% of instances required human adjudication after one prompt refinement iteration. The disagreements that did occur fell into clear thematic categories, which we detail below to illustrate the models' confusions:

Disagreement Themes (Structured Prompt, Personalized Service):

Theme 1: Recognition of Explicit Personalization Keywords (11 cases) – One model sometimes missed obvious personalization terms that the other caught.
Examples: LLM1 failed to recognize *"custom clothing company"* as implying personalization, whereas LLM2 did. In another case, LLM2 correctly flagged *"Custom"* in a phrase ("Custom Comfort Casual line") while LLM1 overlooked it. Such misses were rare but pointed to differences in keyword sensitivity.

Theme 2: Implicit vs. Explicit Personalization (8 cases) – One model required direct mentions of personalization, while the other inferred it from context.
Examples: "creating his look" – LLM1 saw this as indicating a personalized service (styling a specific individual), but LLM2 argued it could be generic. *"fit wasn't right for him off the rack"* – LLM2 recognized this implies a need for personalization, while LLM1 focused only on the fit issue without drawing the personalization inference. *"put your own twist on it"* – LLM1 interpreted this as an invitation to customize, whereas LLM2 wanted a more explicit term.

Theme 3: General Product Features vs. Individual Customization (3 cases) – Confusion over whether describing product suitability for certain customers counts as personalization.

Peer Review Version

Examples: "great for clients that run warm" – LLM1 argued this shows adaptation to a client type (personalization), but LLM2 correctly noted it's just product positioning (a feature for a segment, not individual tailoring). *"better fit"* – LLM2 took this as personalization, but it was in context a general quality claim. *"lifestyle" references* – LLM1 sometimes claimed a generic lifestyle mention was adaptation, whereas LLM2 treated it as describing a target demographic.

After prompt clarification (e.g., emphasizing that broad targeting of segments isn't individual personalization), these discrepancies were resolved.

Summary of Structured-Prompt Performance: Overall, the structured STAMP prompts for Study 1 construct yielded high inter-LLM reliability (α =.75–.82) and low disagreement rates, matching the reliability of human coders. The careful definitions and examples helped the models agree on nuanced content (e.g., what counts as "timeless style" or "personalized service"). In contrast, a definition-only prompt baseline (providing only a brief construct definition and the task) resulted in significantly lower agreement: for the personalized-service construct, the models' α = .40–.55 with frequent divergent interpretations. We analyzed those baseline disagreements as well, to highlight what the structured prompt added:

For the full structured prompt for Classic Luxury Style (Study 1), Recommendation (Study 2), and Value Confirmation (Study 3), see the following link

https://figshare.com/s/b4eb9b11d629f77cfadf

For the datasets, see the following link

https://figshare.com/s/39b839ee953d8bdc8650

Peer Review Version

Web Appendix G: STAMP Workflow from Definition-Only to Full STAMP Prompt (Personalized Service (Study1))

The Development Process

Step 1: Initial Classification with Definition-Only Prompt.

Researchers begin with a basic prompt that contains only the definition of the construct. When two LLMs classify the same text using this simple prompt, they often disagree due to ambiguities and edge cases.

Step 2: Disagreement Analysis

A third LLM analyzes the disagreements and categorizes them into thematic patterns. In the example, six major disagreement themes emerged for the "Personalized_service_products" construct:

1. **Generic vs. Personalized Language** (28 cases) - LLMs disagreed on whether general business phrases constitute actual personalization
2. **Empty/Missing Content** (15 cases) - How to handle statements with no substantive content
3. **Product Features vs. Customization** (12 cases) - Whether describing product features equals personalization
4. **Consultation Offers vs. Actual Personalization** (8 cases) - Whether offering consultations constitutes personalized service
5. **Customer Choice vs. Brand Customization** (6 cases) - Whether customer selection from options equals personalization
6. **Incomplete Analysis** (4 cases) - Template or incomplete responses from one LLM

Step 3: Converting Disagreements into Criteria

Each disagreement theme directly informs specific inclusion and exclusion criteria:

Inclusion Criteria target the positive cases that were correctly identified:

- **Tailor-made offering** (addresses themes 1, 3, 5)
- **Explicitly personalized experience** (addresses themes 4, 6)
- **Emphasis on individual fit** (addresses themes 1, 3)

Exclusion Criteria target the false positives that caused disagreement:

- **Standard retail variety** (addresses theme 3 - product features confusion)
- **Generic invite without personalization** (addresses theme 4 - consultation confusion)
- **Non-text content only** (addresses theme 2 - empty content handling)

Example Selection

The disagreement cases provide concrete examples for the prompt. Edge cases that caused confusion become negative examples with explanations, while clear positive cases become positive examples.

Peer Review Version

STAMP output

Definition-only Prompt

You are a text classifier to identify the tactic "Personalized_service_products": content showing that a brand customizes products or services to an individual client's needs, desire, lifestyle, or preferences.

You will be provided with two pieces of information: <context> and <statement>. Your task is to determine if the "statement" qualifies as the tactic.

Please analyze the given statement and provide your output in the following JSON format:

{
"reasoning": "[criteria]; explanation: [Your step by step reasoning, highlighting keywords]",
"tactic_prediction": [0 or 1]
}

Disagreement Themes (Definition-Only Prompt, Personalized Service):

Using only a basic instruction (e.g., *"Decide if the statement shows the brand customizing its products or services to the customer's needs"*), the models often disagreed due to lack of guidance:

Theme 1: Generic vs. Personalized Language (28 cases) - One model marked posts as personalized based on generic positive language, while the other required explicit customization cues. E.g., *"If you're like me..."* – LLM2 mistakenly took a conversational opener as personalization, whereas LLM1 correctly did not. Both models lacked a clear rule here without the structured prompt.

Theme 2: Handling Empty/Missing Content (15 cases) - Without specified criteria, one model might try to categorize even irrelevant input. E.g., an input that was just "!" – LLM1 correctly refused to classify (no content), but LLM2 *"hallucinated"* finding words like "customizing" and erroneously made a judgment. The structured prompt explicitly prevented this by defining non-text exclusion.

Theme 3: Product Features vs. Customization (12 cases) - The definition-only prompt left ambiguity on whether product-feature mentions imply personalization. LLM2 often assumed *any* product feature (like "well-fitted suit") signaled personal fit; LLM1 generally did not. Without explicit criteria, they diverged on these borderline cases.

Theme 4: Consultation Offers vs. Actual Personalization (8 cases) - Phrases like "Contact us for a consultation" caused confusion. One model (LLM2) tended to count any consultation offer as personalization, whereas the other (LLM1) was unsure. The structured prompt's exclusion criterion clarified that generic invitations alone do not count.

Peer Review Version

Theme 5: Customer Choice vs. Brand Customization (6 cases) - The models disagreed if a post said a client *chose* something vs. the brand *custom-made* something. LLM2 sometimes conflated customer choice (e.g., "client picked out a style") with personalization; LLM1 did not. Clear guidance was needed to distinguish these (which the structured prompt provided by focusing on brand actions).

Theme 6: Incomplete Analysis or Template Responses (4 cases) - The definition-only setting led to occasional failures like one model giving a templated or no answer (e.g., returning an error code or a placeholder). The structured approach with a fixed JSON format mitigated such cases by enforcing a response structure and rationale.

These issues illustrate how the multi-part STAMP prompt (construct definitions + criteria + examples) drove substantial improvements in reliability and interpretative alignment between models. Under the full STAMP prompt, many of the above ambiguities were resolved via explicit instructions—resulting in the high agreement reported for Study 1 in the main text.

1. Interpretation of Generic vs. Personalized Language (Count: 28)
AI1 and AI2 disagree on whether general business language constitutes personalization.

Sample statements:

"AI1 correctly identifies that acknowledging someone's current situation (being in gym shorts) does not constitute actual customization of products or services to meet specific client needs"
"AI2 answer incorrectly interprets the casual phrase 'If you're like me' as evidence of personalized service, when this is simply a common conversational opener"
"AI2 answer incorrectly interprets 'fresh new looks in that closet' as evidence of personalization, when this is actually a common generic marketing phrase"
2. Analysis of Empty/Missing Content (Count: 15)

Disagreement over how to handle statements with no actual content.

Sample statements:

"AI1 answer correctly identifies that both the context and statement are empty (containing only '!') and appropriately concludes that classification is impossible"
"AI2 answer incorrectly claims to find keywords like 'customizing' and 'individual's needs' in the statement, which clearly do not exist"
"AI2 appears to be hallucinating content that is not present in the actual input"
3. Product Features vs. Customization (Count: 12)

Disagreement on whether describing product features equals personalization.

Sample statements:

"AI1 correctly identifies that the statement lacks any mention of customization or personalization - it simply lists generic product features"

Peer Review Version

"AI2 answer incorrectly assumes that mentioning specific product features automatically indicates personalization"
"AI1 correctly identifies that 'well-fitted suit' is merely a general statement about suit quality rather than a description of personalized services"
4. Consultation/Appointment Offers vs. Actual Personalization (Count: 8)

Disagreement on whether offering consultations constitutes personalized service.

Sample statements:

"AI1 correctly identifies that the statement only provides contact information and does not contain explicit evidence of customization"
"AI2 answer incorrectly assumes that offering consultation automatically equals personalized service"
"The word 'consultation' alone does not inherently indicate customization - it could be a standard consultation process"
5. Customer Choice vs. Brand Customization (Count: 6)

Disagreement on whether customer selection from options equals personalization.

Sample statements:

"AI1 correctly identifies that simply choosing from an existing product line does not constitute personalization"
"AI2 answer incorrectly interprets a standard product selection as personalization, confusing customer choice with actual customization"
"AI2 incorrectly conflates customer choice with personalized service, misinterpreting 'client picked out' as evidence of brand customization"
6. Incomplete Analysis/Template Responses (Count: 4)

Disagreement due to one AI providing incomplete or template responses.

Sample statements:

"AI1 answer provides a template-like response with placeholder text and gives a conditional final answer without actually analyzing any content"
"AI1 answer provides a structured framework but fails to recognize the absence of actual content"
"AI1 answer encountered an API error and returned -1, providing no classification analysis"When moving from the Definition-only Prompt to the STAMP Prompt

Researchers can systematically utilize disagreement themes and cases from the "Definition only prompt" to develop a comprehensive STAMP prompt through a structured, iterative process. The definition prompt can serve as a theory exploration for new constructs using the LLM's vast knowledge and assumptions.

Peer Review Version

Web Appendix H: Robustness of Results Across Models.

To enhance the robustness of our findings, we used bootstrap resampling (1,000 iterations) with temperature = 0 to produce deterministic output, rather than the default temperature of 1 used in ChatGPT for the main study. The results show that GPT-4.1 and Gemini 2.0 Flash perform indistinguishably from each other and outperform human experts in both reliability (alpha) and predictive validity (precision, recall, F1). We therefore focus on these two models in the main paper.

Table W3. Inter-rater Reliability Between LLMs and Human Coders with Bootstrap Confidence Intervals

Construct & Prompt	Krippendorff's α (GPT-4.1 and Flash 2.0) [95% CI]	Disagreement Rate (GPT-4.1 and Flash 2.0)	Krippendorff's α (Human Expert / Crowd) [95% CI]	Disagreement Rate (Human Expert / Crowd)	Krippendorff's α (Trained RA / Crowd) [95% CI]	Disagreement Rate (Trained RA / Crowd)
Classic Luxury Style – Definition-Only	.42 [.30, .53]	6.41%	N/A	N/A	N/A	N/A
Classic Luxury Style – STAMP Prompt	.73 [.65, .80]	4.30%	.80 [.72, .87] ✓	2.80%	.69 [.60, .78]	4.00%
Personalized Service – Definition-Only	.67 [.60, .74]	7.88%	N/A	N/A	N/A	N/A
Personalized Service – STAMP Prompt	.86 [.81, .92] ✓	2.30%	.90 [.84, .94] ✓	1.50%	.79 [.71, .86] ✓	3.00%
Recommendation – Definition-Only	.46 [.38, .53]	23.87%	N/A	N/A	N/A	N/A
Recommendation – STAMP Prompt	.72 [.66, .78]	13.00%	.70 [.65, .76]	14.00%	.47 [.39, .54]	24.83%
Value Confirmation – Definition-Only	.67 [.60, .72]	10.64%	N/A	N/A	N/A	N/A
Value Confirmation – STAMP Prompt	.81 [.74, .87] ✓	2.91%	.75 [.67, .82] ✓	3.90%	N/A	N/A

Note: Bootstrap confidence intervals calculated using 1,000 replicates with percentile method (2.5th and 97.5th percentiles). ✓ indicates cells where the lower bound of the 95% CI meets or exceeds the 0.67 threshold, demonstrating statistical confidence in achieving acceptable reliability (Krippendorff, 2004). Human Expert / Crowd comparison is used as the primary benchmark for human-human performance. RA denotes Research Assistant. Flash 2.0 refers to Gemini 2.0 Flash.

Peer Review Version

Table W4. STAMP Classification Performance Metrics with Bootstrap Confidence Intervals

Model / Rater	Metric	Classic Luxury Style	Personalized Service	Recommendation	Value Confirmation
GPT-4.1 (Flagship AI)	F1	.83 [.77, .89] ✓	.91 [.87, .96] ✓	.91 [.89, .93] ✓	0.83 [0.76, 0.89] ✓
	Precision	.82 [.73, .90]	.93 [.87, .98]	.89 [.86, .92]	0.78 [0.69, 0.87]
	Recall	.84 [.77, .91]	.90 [.84, .96]	.94 [.91, .96]	0.89 [0.82, 0.95]
Claude 4.5 Sonnet (Flagship AI)	F1	.81 [.75, .87] ✓	.90 [.85, .94] ✓	.87 [.84, .90] ✓	0.84 [0.76, 0.90] ✓
	Precision	.75 [.67, .84]	.83 [.76, .91]	.89 [.86, .92]	0.89 [0.81, 0.96]
	Recall	.87 [.80, .94]	.98 [.94, 1.0]	.85 [.82, .89]	0.80 [0.70, 0.89]
Gemini 2.0 Flash (Low-cost AI)	F1	.82 [.76, .88] ✓	.88 [.83, .93] ✓	.88 [.86, .90] ✓	0.81 [0.74, 0.88] ✓
	Precision	.82 [.73, .89]	.81 [.72, .88]	.83 [.79, .86]	0.78 [0.68, 0.87]
	Recall	.83 [.75, .90]	.98 [.94, 1.0]	.94 [.92, .97]	0.85 [0.76, 0.93]
Gemini 2.5 Flash (Low-cost AI)	F1	.76 [.68, .83]	.88 [.82, .93] ✓	.86 [.84, .89] ✓	0.75 [0.66, 0.83]
	Precision	.78 [.68, .87]	.80 [.72, .87]	.80 [.76, .84]	0.83 [0.73, 0.92]
	Recall	.75 [.65, .83]	.98 [.94, 1.0]	.94 [.91, .96]	0.68 [0.56, 0.79]
Claude Haiku 3.5 (Low-cost AI)	F1	.81 [.73, .86] ✓	.86 [.80, .90] ✓	.84 [.82, .87] ✓	0.72 [0.63, 0.79]
	Precision	**.72 [.64, .80]**	**.79 [.70, .86]**	**.74 [.70, .79]**	0.63 [0.53, 0.73]
	Recall	.91 [.84, .96]	.94 [.89, .99]	.98 [.96, .99]	0.84 [0.75, 0.92]
GPT-4.1-mini (Low-cost AI)	F1	.61 [.50, .71]	.84 [.76, .89] ✓	.85 [.82, .88] ✓	0.69 [0.59, 0.77]
	Precision	.98 [.92, 1.0]	.95 [.90, 1.0]	.94 [.91, .96]	0.73 [0.61, 0.84]
	Recall	**.45 [.34, .55]**	**.74 [.64, .83]**	**.78 [.74, .83]**	0.64 [0.53, 0.75]
Crowd (Modal) (Human Benchmark)	F1	.82 [.75, .88] ✓	.91 [.85, .95] ✓	.89 [.86, .91] ✓	0.77 [0.69, 0.83]
	Precision	.94 [.88, .99]	.94 [.87, .98]	.84 [.80, .87]	0.68 [0.58, 0.77]
	Recall	.72 [.63, .82]	.88 [.80, .95]	.94 [.92, .97]	0.89 [0.81, 0.96]
Trained RA (Human Benchmark)	F1	.76 [.68, .83]	.86 [.79, .91] ✓	.86 [.83, .89] ✓	N/A
	Precision	.84 [.75, .92]	.88 [.81, .95]	.84 [.81, .88]	N/A
	Recall	.70 [.60, .79]	.83 [.75, .91]	.88 [.85, .91]	N/A

Note: Bootstrap confidence intervals calculated using 1,000 replicates with percentile method (2.5th and 97.5th percentiles). F1 ≥ 0.80 (good performance). ✓ = F1 ≥ 0.80. Bold indicates problematic precision or recall. The STAMP is revised one round by the researcher. The RA were trained for one to two rounds.

Peer Review Version

References

Anthropic (2025), "Use XML tags to structure your prompts," (accessed March 13, 2025), [available at https://docs.anthropic.com/en/docs/build-with-claude/prompt-engineering/use-xml-tags].

Bermejo, Victor J., Alberto Gago, Ramiro H. Gálvez, and Nimrod Harari (2025), "LLMs outperform outsourced human coders on complex textual analysis," *Scientific Reports*, 15, 40122. https://doi.org/10.1038/s41598-025-23798-y.

Bojić, Ljubiša, Olena Zagovora, Agne Zelenkauskaite, Viktor Vuković, Milica Čabarkapa, Sanja Veseljević Jerković, and Aleksandra Jovančević (2025), "Comparing large language models and human annotators in latent content analysis of sentiment, political leaning, emotional intensity and sarcasm," *Scientific Reports*, 15 (1), 11477. https://doi.org/10.1038/s41598-025-96508-3.

Carlson, Nora A. and Vanessa Burbano (2025), "The use of LLMs to annotate data in management research: Foundational guidelines and warnings," *Strategic Management Journal*, 46 (forthcoming). https://doi.org/10.1002/smj.70023.

Center for Research on Foundation Models (2025), "Holistic evaluation of language models (HELM): MMLU," (accessed October 13, 2025), [available at https://crfm.stanford.edu/helm/mmlu/latest/].

Chae, Y. and T. Davidson (2025), "Large language models for text classification: From zero-shot learning to instruction-tuning," *Sociological Methods & Research*, forthcoming. https://doi.org/10.1177/00491241251325243.

Chakraborty, Ishita, Kyu Chiong, Harry Dover, and K. Sudhir (2025), "Can AI and AI-hybrids detect persuasion skills? Salesforce hiring with conversational video interviews," *Marketing Science*, 44 (1), 30–53. https://doi.org/10.1287/mksc.2023.0149.

Cronbach, L.J. and P.E. Meehl (1955), "Construct validity in psychological tests," *Psychological Bulletin*, 52 (4), 281–302.

Gilardi, Fabrizio, Meysam Alizadeh, and Maël Kubli (2023), "ChatGPT outperforms crowd workers for text-annotation tasks," *Proceedings of the National Academy of Sciences*, 120 (30), e2305016120. https://doi.org/10.1073/pnas.2305016120.

Goli, A. and A. Singh (2024), "Frontiers: Can large language models capture human preferences?" *Marketing Science*, 43 (4), 709–22.

Google Cloud (2025), "Structure prompts | Generative AI," (accessed September 24, 2025), [available at https://cloud.google.com/vertex-ai/generative-ai/docs/learn/prompts/structure-prompts].

Peer Review Version

Hartmann, J., M. Heitmann, C. Siebert, and C. Schamp (2023), "More than a feeling: Accuracy and application of sentiment analysis," *International Journal of Research in Marketing*, 40 (1), 75–87.

Hendrycks, D., C. Burns, S. Basart, A. Zou, M. Mazeika, D. Song, and J. Steinhardt (2021), "Measuring massive multitask language understanding," working paper, arXiv:2009.03300.

Humphreys, A. and R.J.H. Wang (2018), "Automated text analysis for consumer research," *Journal of Consumer Research*, 44 (6), 1274–1306.

Krippendorff, Klaus (2018), *Content Analysis: An Introduction to Its Methodology*, 3rd ed. Thousand Oaks, CA: Sage Publications.

Ludwig, S., K. De Ruyter, M. Friedman, E.C. Brüggen, M. Wetzels, and G. Pfann (2013), "More than words: The influence of affective content and linguistic style matches in online reviews on conversion rates," *Journal of Marketing*, 77 (January), 87–103.

Martin, A. and T. Kushwaha (2024), "Can words speak louder than actions? Using top management teams' language to predict myopic marketing spending," *Journal of Marketing*, forthcoming. https://doi.org/10.1177/00222429241244804.

Netzer, O., R. Feldman, J. Goldenberg, and M. Fresko (2012), "Mine your own business: Market-structure surveillance through text mining," *Marketing Science*, 31 (3), 521–43.

Short, J.C., J.C. Broberg, C.C. Cogliser, and K.H. Brigham (2010), "Construct validation using computer-aided text analysis (CATA): An illustration using entrepreneurial orientation," *Organizational Research Methods*, 13 (2), 320–47.

Timoshenko, A. and J.R. Hauser (2019), "Identifying customer needs from user-generated content," *Marketing Science*, 38 (1), 1–20.

Törnberg, P. (2025), "Large language models outperform expert coders and supervised classifiers at annotating political social media messages," *Social Science Computer Review*, 43 (6), 1181–95. https://doi.org/10.1177/08944393241286471.

Villarroel Ordenes, F., G. Packard, J. Hartmann, and D. Proserpio (2025), "Using traditional text analysis and large language models in service failure and recovery," *Journal of Service Research*, 28 (3), 382–88.

Wei, J., X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E.H. Chi, Q.V. Le, and D. Zhou (2022), "Chain-of-thought prompting elicits reasoning in large language models," *Advances in Neural Information Processing Systems*, 35, 24824–837.

Yeomans, M. (2021), "A concrete example of construct construction in natural language," *Organizational Behavior and Human Decision Processes*, 162, 81–94.

Peer Review Version